

TACHS Practice Exam - Exam 3 - Part A
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Exam Part: A
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QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R3-S01
Section: Reading
Stimulus Type: Literary Passage
Title: The Quiet Bell

Text:

Mara had volunteered to restore the brass bell from the town's abandoned ferry landing. She imagined polishing away a little tarnish and finishing before lunch. Instead, she found a cracked wooden handle, layers of green corrosion, and a clapper frozen in place.

At the workshop, Mr. Ibarra showed her how to loosen each part without damaging it. Mara wanted to hurry, but every shortcut left a scratch or bent an edge. She began keeping notes: which solution worked, how long each piece soaked, and where every washer belonged.

On the morning of the river festival, Mara fastened the repaired bell beside the new walking path. The mayor invited her to ring it. She pulled the handle, but only a dull knock sounded. Her face warmed as the crowd waited. Then Mara noticed that the clapper's leather strap had twisted during installation. She calmly removed the pin, straightened the strap, and tried again.

The clear note traveled over the water. Mara smiled, not because the bell sounded perfect, but because she now knew what to do when it did not.

STIMULUS END

TACHS-R3-001

Section: Reading
Type: mcq
Category: Reading
Skill: Main Idea, Central Thought, or Summary
Stimulus ID: TACHS-R3-S01
Prompt:

Which statement best expresses the central idea of the passage?

- A) Careful work teaches Mara how to solve an unexpected problem.
- B) Public festivals should preserve objects from local history.
- C) Mr. Ibarra performs the most difficult part of the restoration.
- D) Old machinery usually works poorly after being repaired.

TACHS-R3-002

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S01

Prompt:

What caused the bell to make only a dull knock?

- A) The clapper was still corroded.
- B) A leather strap had become twisted.
- C) The wooden handle cracked again.
- D) A washer had been left at the workshop.

TACHS-R3-003

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S01

Prompt:

What can reasonably be inferred about Mara at the end of the passage?

- A) She regrets volunteering for the project.
- B) She believes the crowd did not notice the delay.
- C) She has become more patient and resourceful.
- D) She plans to replace the bell with a new one.

TACHS-R3-004

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R3-S01

Prompt:

How does the final sentence contribute to the passage?

- A) It introduces a new conflict about the bell's condition.
- B) It explains why the festival was held beside the river.
- C) It reveals that Mara cares only about pleasing the crowd.
- D) It emphasizes that learning to respond to setbacks matters more than perfection.

STIMULUS START

Stimulus ID: TACHS-R3-S02

Section: Reading

Stimulus Type: Informational Passage

Title: A Garden That Holds Rain

Text:

After a heavy storm, water rushing from roofs and pavement can overwhelm drains and carry soil or litter into streams. A rain garden offers a small-scale solution. It is a shallow planted area placed where runoff naturally collects. Rather than sending water immediately into a pipe, the garden briefly holds it.

A rain garden is not a permanent pond. Its soil is prepared to absorb water, and its surface usually dries within a day or two. Deep-rooted native plants are often suitable because their roots create channels in the soil. These plants may also require less watering after they become established.

Location matters. Builders avoid placing rain gardens directly against foundations or over underground utilities. They also test how quickly the soil drains. If water remains too long, mosquitoes may breed and plant roots may decay.

One garden cannot prevent every flood. However, many small gardens distributed through a neighborhood can reduce the amount of runoff reaching storm drains at the same moment. They can also provide food and shelter for insects and birds. Thus, a feature designed to manage water may improve local habitat as well.

STIMULUS END

TACHS-R3-005

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S02

Prompt:

According to the passage, why are deep-rooted plants useful in rain gardens?

- A) Their roots form channels that help water enter the soil.
- B) Their leaves prevent all rain from reaching the ground.
- C) Their flowers remove litter from nearby streams.
- D) Their stems strengthen building foundations.

TACHS-R3-006

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S02

Prompt:

As used in the final paragraph, “distributed” most nearly means

- A) sold for a profit
- B) spread among different places
- C) measured with precision
- D) removed from public land

TACHS-R3-007

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S02

Prompt:

The author's primary purpose is to

- A) argue that storm drains should be eliminated
- B) describe the history of neighborhood landscaping
- C) explain how rain gardens work and why proper design matters
- D) compare the appearance of several native plants

TACHS-R3-008

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S02

Prompt:

Which conclusion is best supported by the passage?

- A) Every yard has soil suitable for a rain garden.
- B) Rain gardens remain filled with water throughout summer.
- C) A single rain garden can protect an entire city from flooding.
- D) The combined effect of many modest projects can be significant.

STIMULUS START

Stimulus ID: TACHS-R3-S03

Section: Reading

Stimulus Type: Humanities Passage

Title: Reading a Building

Text:

People often treat buildings as backgrounds, noticing them only when searching for an entrance. Yet a building can be read as evidence of the choices and needs of the people who made it. A wide staircase may announce a public gathering place. A row of identical windows may reveal that builders valued efficiency, while carved stone above a doorway may show where they chose to spend extra labor.

Materials provide clues, too. Brick made from nearby clay can connect a structure to its region. A later wall built with different brick may mark an expansion. Even repairs tell a story: a patch can show both where damage occurred and what materials were available when it was fixed.

Interpreting these clues requires caution. A modern visitor might assume that a tiny window was intended for decoration, although it may have reduced heat loss. One feature can also serve several purposes. An awning can shelter a doorway, shade a room, and advertise a shop.

Looking closely does not mean every guess will be correct. It means replacing a quick impression with useful questions: What problem did this feature solve? Which parts are original? What changed over time? Buildings become more than scenery when observers recognize them as records shaped by climate, technology, money, and human intention.

STIMULUS END

TACHS-R3-009

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R3-S03

Prompt:

Which sentence best summarizes the passage?

- A) Architectural features can provide evidence, but observers must interpret them thoughtfully.
- B) Old buildings should always be restored with local materials.
- C) Modern buildings contain fewer meaningful details than older ones.
- D) Decorative features are the most reliable clues to a building's purpose.

TACHS-R3-010

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S03

Prompt:

In the first paragraph, "announce" most nearly means

- A) publish
- B) shout
- C) signal
- D) celebrate

TACHS-R3-011

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S03

Prompt:

What might different brick in a later wall indicate?

- A) The building was expanded.
- B) The original clay was too colorful.
- C) The windows were once larger.
- D) The structure was moved to another region.

TACHS-R3-012

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S03

Prompt:

Why does the author mention that an awning can serve several purposes?

- A) To prove that shop signs are unnecessary
- B) To warn that a feature may not have one simple explanation
- C) To show that awnings are more useful than windows
- D) To recommend adding awnings to historic buildings

TACHS-R3-013

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S03

Prompt:

Which word best describes the author's tone?

- A) Alarmed
- B) Amused
- C) Doubtful
- D) Curious

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE3-S01

Section: Written Expression

Stimulus Type: Revision Passage

Title: Borrowing More Than Books

Text:

[1] Our school library recently began lending small tool kits. [2] Each kit contains basic items such as a screwdriver, measuring tape, and level. [3] Students may borrow one for three days with a library card. [4] The idea began after a survey showed that many families occasionally needed tools but did not want to purchase them. [5] The kits were popular right away. [6] During the first month, every kit was borrowed at least twice. [7] The library also displays novels near the front desk. [8] Because the program is useful, the student council plans to collect donations for additional kits.

STIMULUS END

TACHS-WE3-001

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S01

Prompt:

Which sentence should be removed because it does not support the passage's focus?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 7
- D) Sentence 8

TACHS-WE3-002

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S01

Prompt:

Which revision most effectively combines sentences 5 and 6?

- A) The kits were popular right away, during the first month.
- B) The kits were immediately popular; every kit was borrowed at least twice during the first month.
- C) Popular right away, and every kit was borrowed at least twice during the first month.
- D) Every kit was borrowed, the kits were popular at least twice during the first month.

TACHS-WE3-003

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S01

Prompt:

Which word best replaces “useful” in sentence 8 to make the statement more precise?

- A) successful
- B) nice
- C) thing-like
- D) okay

TACHS-WE3-004

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S01

Prompt:

Which sentence would provide the most effective conclusion?

- A) Libraries have existed for a very long time.
- B) Some screwdrivers have flat tips, while others have crossed tips.
- C) Surveys can contain questions with several answer choices.
- D) With more kits available, even more families can complete occasional repairs without buying rarely used tools.

STIMULUS START

Stimulus ID: TACHS-WE3-S02

Section: Written Expression

Stimulus Type: Revision Passage

Title: Preparing for the Night Walk

Text:

[1] Next Friday, the science club will lead an evening nature walk in Pine Hollow Park. [2] Participants should bring a flashlight and wear sturdy shoes. [3] The group will pause at three marked locations to listen for frogs and insects. [4] Club members has practiced identifying each species by its call. [5] Before the walk begins, a park educator will explain how artificial light can affect nocturnal animals. [6] Participants will then cover their flashlights with red filters, which are less disruptive than bright white beams. [7] Everyone must remain on the trail, speak quietly, and to follow the leader. [8] These steps will help the group observe wildlife responsibly.

STIMULUS END

TACHS-WE3-005

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE3-S02

Prompt:

Which revision corrects sentence 4?

- A) Club members have practiced identifying each species by its call.
- B) Club members having practiced identifying each species by its call.
- C) Club members has been practicing to identify each species by its call.
- D) Club members is practiced identifying each species by its call.

TACHS-WE3-006

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S02

Prompt:

Which revision makes the series in sentence 7 parallel?

- A) Everyone must remain on the trail, speaking quietly, and follows the leader.
- B) Everyone must remain on the trail, quiet speech, and following the leader.
- C) Everyone must remain on the trail, speak quietly, and follow the leader.
- D) Everyone must remaining on the trail, speak quietly, and to follow the leader.

TACHS-WE3-007

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S02

Prompt:

Which revision of sentence 3 is most concise without changing its meaning?

- A) The group, which is the group participating, will make pauses at three locations that are marked.
- B) At three marked locations, there will be pauses made by the group to listen.
- C) The group will pause and stop at three marked locations for listening purposes.

D) The group will stop at three marked locations to listen for frogs and insects.

TACHS-WE3-008

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S02

Prompt:

Where should this sentence be placed?

“This brief lesson will prepare walkers to limit their impact on the park.”

A) After sentence 2

B) After sentence 5

C) After sentence 7

D) After sentence 8

TACHS-WE3-009

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

A) Neither of the two routes lead directly to the museum.

B) The box of donated maps is beside the entrance.

C) Each of the volunteers were given a name tag.

D) One of those lanterns need a new battery.

TACHS-WE3-010

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: None

Prompt:

Which choice is a complete sentence?

A) Although the rehearsal ended later than expected.

B) Carrying the painted scenery into the auditorium.

C) Because the musicians needed one final practice.

D) The stage manager checked every microphone before the doors opened.

TACHS-WE3-011

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: None

Prompt:

Which revision is clearest?

“When Lena spoke with Ava, she said the schedule had changed.”

- A) When Lena spoke with Ava, she mentioned it.
- B) Lena told Ava that the schedule had changed.
- C) Speaking with Ava, the schedule had changed according to Lena.
- D) She told her about the changed schedule when they spoke.

TACHS-WE3-012

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The robotics team tested its new design on Tuesday.

Line 2: Every wheel turned smoothly during the first trial.

Line 3: Later, the students adjusted the sensors careful.

Line 4: They recorded the results in a shared notebook.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE3-013

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: Maya and me presented the proposal to the committee.

Line 2: We explained how the new signs would help visitors.

Line 3: Several members asked thoughtful questions.

Line 4: The committee approved the plan unanimously.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M3-001

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is $\frac{3}{5}$ of 45?

A) 15

B) 27

C) 30

D) 75

TACHS-M3-002

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A jacket priced at \$64 is discounted by 25%. What is the sale price?

A) \$16

B) \$48

C) \$39

D) \$56

TACHS-M3-003

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which value is greatest?

A) 0.62

B) 61%

C) $\frac{5}{8}$

D) 0.6

TACHS-M3-004

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $4x - 7 = 21$.

A) 7

B) 14

C) 16

D) 28

TACHS-M3-005

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The sequence begins 6, 11, 16, 21. Which expression gives the n th term?

- A) $5n - 1$
- B) $5n + 1$
- C) $6n - 1$
- D) $6n$

TACHS-M3-006

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

If $y = 3x + 2$, what is y when $x = -4$?

- A) -14
- B) -12
- C) -10
- D) 10

TACHS-M3-007

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The numbers of pages read on five days were 18, 22, 15, 25, and 20. What was the mean?

- A) 20
- B) 18
- C) 21
- D) 25

TACHS-M3-008

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A bag contains 4 blue, 3 green, and 5 yellow tiles. One tile is selected, replaced, and then another is selected. What is the probability that both tiles are green?

- A) $\frac{1}{16}$
- B) $\frac{1}{12}$
- C) $\frac{1}{9}$
- D) $\frac{1}{4}$

TACHS-M3-009

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A triangle has angles measuring 48° and 67° . What is the measure of the third angle?

- A) 55°
- B) 60°
- C) 65°
- D) 75°

TACHS-M3-010

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangle has area 84 square centimeters and length 12 centimeters. What is its perimeter?

- A) 19 cm
- B) 42 cm
- C) 84 cm
- D) 38 cm

TACHS-M3-011

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A recipe uses 750 milliliters of broth. How many liters is this?

- A) 0.075 L
- B) 7.5 L
- C) 75 L
- D) 0.75 L

TACHS-M3-012

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A bus leaves at 9:35 a.m. and arrives at 12:10 p.m. How long is the trip?

- A) 2 hours 25 minutes
- B) 3 hours 25 minutes
- C) 3 hours 35 minutes
- D) 2 hours 35 minutes

SECTION 4 - ABILITY

TACHS-A3-001

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: ○

Top-right: ○○

Bottom-left: ●

Bottom-right: ?

Across each row, one matching shape is added. Down each column, open shapes become filled.

- A) ●●
- B) ○●
- C) ■■
- D) ●●●

TACHS-A3-002

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: ↑

Top-right: →

Bottom-left: ←

Bottom-right: ?

Across each row, the arrow rotates 90° clockwise.

- A) →
- B) ↓
- C) ↑
- D) ←

TACHS-A3-003

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ○ | □ | ○□

Row 2: △ | ◇ | △◇

Row 3: ☆ | + | ?

In each row, the third cell combines the first two cells in the same order.

A) +☆

B) ☆+

C) ★+

D) ☆×

TACHS-A3-004

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: open shape ○ above filled shape ■

Top-right: filled shape ■ above open shape ○

Bottom-left: open shape △ above filled shape ◆

Bottom-right: ?

Across each row, the top and bottom positions switch.

A) open △ above filled ◆

B) filled ▲ above open ◇

C) open ◇ above filled ▲

D) filled ◆ above open △

TACHS-A3-005

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ○ | ● | ○

Row 2: ● | ○ | ●

Row 3: ○ | ● | ?

The cells alternate between open and filled circles across every row and column.

- A) ●
- B) ○
- C) □
- D) ■

TACHS-A3-006

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded in half from left to right. One hole is punched near the upper-right corner of the folded rectangle. After unfolding, where are the holes?

- A) Near both lower corners
- B) At the centers of the left and right edges
- C) Near both upper corners
- D) One near the upper-right corner only

TACHS-A3-007

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded bottom to top, then left to right. One hole is punched at the exact center of the final folded square. After both folds are undone, what results?

- A) Two holes along the vertical centerline
- B) Four holes, one in each quadrant, symmetric across both centerlines
- C) Four holes along the top edge
- D) Eight holes around the border

TACHS-A3-008

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular paper is folded in half from top to bottom. A triangular notch is cut from the folded edge, away from both corners. What appears after unfolding?

- A) A diamond-shaped opening centered on the horizontal centerline
- B) Two separate notches, one on each outer edge
- C) One triangular notch on the top edge
- D) A circular opening at the center

TACHS-A3-009

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded along the diagonal from upper-left to lower-right. One hole is punched in the folded triangle away from the diagonal and all edges. After unfolding, what results?

- A) Two holes reflected across that diagonal
- B) Two holes reflected across the vertical centerline
- C) Four holes reflected across both centerlines
- D) One hole on the diagonal

TACHS-A3-010

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The figures ○●○, □■□, and △▲△ share a rule. Which choice follows the same rule?

- A) ◆◆◆
- B) ★☆☆
- C) ◇◆◆
- D) ●○○

TACHS-A3-011

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The arrangements “○ inside □,” “△ inside ◇,” and “☆ inside ○” share a rule. Which choice follows the same rule?

- A) ■ inside ◆
- B) + inside ×
- C) ● beside □
- D) ◇ inside △

TACHS-A3-012

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The figures $\uparrow\rightarrow\downarrow$, $\rightarrow\downarrow\leftarrow$, and $\downarrow\leftarrow\uparrow$ share a rule. Which choice follows the same rule?

- A) $\uparrow\downarrow\rightarrow$
- B) $\leftarrow\downarrow\rightarrow$
- C) $\rightarrow\leftarrow\downarrow$
- D) $\leftarrow\uparrow\rightarrow$

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R3-001 - Correct: A - Correct Answer: Careful work teaches Mara how to solve an unexpected problem. - Explanation: Mara's patient restoration work gives her the knowledge and confidence to correct the final problem.

TACHS-R3-002 - Correct: B - Correct Answer: A leather strap had become twisted. - Explanation: Mara notices that the clapper's leather strap twisted during installation.

TACHS-R3-003 - Correct: C - Correct Answer: She has become more patient and resourceful. - Explanation: Mara calmly diagnoses and repairs the problem instead of rushing or giving up.

TACHS-R3-004 - Correct: D - Correct Answer: It emphasizes that learning to respond to setbacks matters more than perfection. - Explanation: The final sentence contrasts a perfect result with Mara's newly developed problem-solving ability.

TACHS-R3-005 - Correct: A - Correct Answer: Their roots form channels that help water enter the soil. - Explanation: The passage directly states that deep roots create channels in the soil.

TACHS-R3-006 - Correct: B - Correct Answer: spread among different places - Explanation: The gardens are located throughout the neighborhood rather than concentrated in one place.

TACHS-R3-007 - Correct: C - Correct Answer: explain how rain gardens work and why proper design matters - Explanation: The passage describes their function, benefits, placement, and drainage requirements.

TACHS-R3-008 - Correct: D - Correct Answer: The combined effect of many modest projects can be significant. - Explanation: Many neighborhood rain gardens together can reduce simultaneous runoff into storm drains.

TACHS-R3-009 - Correct: A - Correct Answer: Architectural features can provide evidence, but observers must interpret them thoughtfully. - Explanation: The passage presents buildings as records while warning readers against quick assumptions.

TACHS-R3-010 - Correct: C - Correct Answer: signal - Explanation: A wide staircase can indicate or signal that the building is a public gathering place.

TACHS-R3-011 - Correct: A - Correct Answer: The building was expanded. - Explanation: The passage says different brick in a later wall may mark an expansion.

TACHS-R3-012 - Correct: B - Correct Answer: To warn that a feature may not have one simple explanation - Explanation: The awning example demonstrates that one architectural feature can serve multiple functions.

TACHS-R3-013 - Correct: D - Correct Answer: Curious - Explanation: The author encourages close observation and thoughtful questions about architectural clues.

WRITTEN EXPRESSION ANSWERS

TACHS-WE3-001 - Correct: C - Correct Answer: Sentence 7 - Explanation: The location of novels does not develop the passage's focus on the tool-lending program.

TACHS-WE3-002 - Correct: B - Correct Answer: The kits were immediately popular; every kit was borrowed at least twice during the first month. - Explanation: The semicolon correctly joins two related independent clauses without creating a run-on.

TACHS-WE3-003 - Correct: A - Correct Answer: successful - Explanation: "Successful" precisely connects the program's demonstrated demand to the plan for expansion.

TACHS-WE3-004 - Correct: D - Correct Answer: With more kits available, even more families can complete occasional repairs without buying rarely used tools. - Explanation: This conclusion reinforces both the planned expansion and the program's central benefit.

TACHS-WE3-005 - Correct: A - Correct Answer: Club members have practiced identifying each species by its call. - Explanation: The plural subject "members" requires the plural helping verb "have."

TACHS-WE3-006 - Correct: C - Correct Answer: Everyone must remain on the trail, speak quietly, and follow the leader. - Explanation: All three items use parallel base-form verbs governed by "must."

TACHS-WE3-007 - Correct: D - Correct Answer: The group will stop at three marked locations to listen for frogs and insects. - Explanation: This version preserves every essential detail without redundancy or awkward passive phrasing.

TACHS-WE3-008 - Correct: B - Correct Answer: After sentence 5 - Explanation: "This brief lesson" clearly refers to the educator's explanation in sentence 5.

TACHS-WE3-009 - Correct: B - Correct Answer: The box of donated maps is beside the entrance. - Explanation: The singular subject "box" correctly agrees with the singular verb "is."

TACHS-WE3-010 - Correct: D - Correct Answer: The stage manager checked every microphone before the doors opened. - Explanation: This choice contains a subject, a complete predicate, and no dependent-word fragment.

TACHS-WE3-011 - Correct: B - Correct Answer: Lena told Ava that the schedule had changed. - Explanation: Naming Lena as the speaker removes the unclear pronoun reference.

TACHS-WE3-012 - Correct: C - Correct Answer: Line 3 - Explanation: The verb "adjusted" requires the adverb "carefully," not the adjective "careful."

TACHS-WE3-013 - Correct: A - Correct Answer: Line 1 - Explanation: The compound subject requires the subject pronoun "I," as in "Maya and I."

MATHEMATICS ANSWERS

TACHS-M3-001 - Correct: B - Correct Answer: 27 - Explanation: $45 \div 5 = 9$, and $9 \times 3 = 27$.

TACHS-M3-002 - Correct: B - Correct Answer: \$48 - Explanation: The discount is $0.25 \times \$64 = \16 , so the price is $\$64 - \$16 = \$48$.

TACHS-M3-003 - Correct: C - Correct Answer: $5/8$ - Explanation: $5/8 = 0.625$, which is greater than 0.62, 0.61, and 0.60.

TACHS-M3-004 - Correct: A - Correct Answer: 7 - Explanation: Add 7 to obtain $4x = 28$, then divide by 4.

TACHS-M3-005 - Correct: B - Correct Answer: $5n + 1$ - Explanation: The sequence increases by 5, and $5(1) + 1$ gives the first term, 6.

TACHS-M3-006 - Correct: C - Correct Answer: -10 - Explanation: Substitute -4: $y = 3(-4) + 2 = -12 + 2 = -10$.

TACHS-M3-007 - Correct: A - Correct Answer: 20 - Explanation: The sum is 100 pages, and $100 \div 5 = 20$.

TACHS-M3-008 - Correct: A - Correct Answer: $1/16$ - Explanation: The probability on each draw is $3/12 = 1/4$, so $1/4 \times 1/4 = 1/16$.

TACHS-M3-009 - Correct: C - Correct Answer: 65° - Explanation: Triangle angles total 180° , and $180^\circ - 48^\circ - 67^\circ = 65^\circ$.

TACHS-M3-010 - Correct: D - Correct Answer: 38 cm - Explanation: The width is $84 \div 12 = 7$, so the perimeter is $2(12 + 7) = 38$.

TACHS-M3-011 - Correct: D - Correct Answer: 0.75 L - Explanation: Since 1,000 milliliters equal 1 liter, $750 \div 1,000 = 0.75$ liter.

TACHS-M3-012 - Correct: D - Correct Answer: 2 hours 35 minutes - Explanation: From 9:35 to 11:35 is two hours, and to 12:10 is 35 more minutes.

ABILITY ANSWERS

TACHS-A3-001 - Correct: A - Correct Answer: ●● - Explanation: Moving right adds one identical shape, while moving down changes each open circle to a filled circle.

TACHS-A3-002 - Correct: C - Correct Answer: ↑ - Explanation: Rotating the bottom-left arrow 90° clockwise changes ← to ↑.

TACHS-A3-003 - Correct: B - Correct Answer: ☆+ - Explanation: Each third cell joins the first and second symbols in their original left-to-right order.

TACHS-A3-004 - Correct: D - Correct Answer: filled ◆ above open Δ - Explanation: The two shapes keep their forms but exchange top and bottom positions across the row.

TACHS-A3-005 - Correct: B - Correct Answer: ○ - Explanation: Alternation requires an open circle after the filled circle in both the final row and column.

TACHS-A3-006 - Correct: C - Correct Answer: Near both upper corners - Explanation: Unfolding reflects the upper-right punch across the vertical centerline to the upper-left.

TACHS-A3-007 - Correct: B - Correct Answer: Four holes, one in each quadrant, symmetric across both centerlines - Explanation: Reversing two perpendicular half-folds reflects the punch across both centerlines, producing four symmetric holes.

TACHS-A3-008 - Correct: A - Correct Answer: A diamond-shaped opening centered on the horizontal centerline - Explanation: A triangular cut on the fold reflects into a matching triangle, forming a diamond-shaped opening.

TACHS-A3-009 - Correct: A - Correct Answer: Two holes reflected across that diagonal - Explanation: Reversing one diagonal fold creates one mirror image across the fold line.

TACHS-A3-010 - Correct: C - Correct Answer: ◇◆◇ - Explanation: Each figure has matching open outer shapes surrounding the filled version of the same shape.

TACHS-A3-011 - Correct: D - Correct Answer: ◇ inside Δ - Explanation: Like every example, this arrangement places one open shape inside a different open shape.

TACHS-A3-012 - Correct: D - Correct Answer: ←↑→ - Explanation: Each arrow turns 90° clockwise from the previous arrow: ← becomes ↑, then →.

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R3-S04

Section: Reading

Stimulus Type: Literary or Narrative Passage

Title: The Washed Grid

Text:

Amaya and Tariq had spent two afternoons drawing a chalk grid across the brick wall behind the community center. The grid would guide a mural of birds crossing a sunrise, and each square corresponded to one square on their smaller sketch. By Friday evening, every line was in place.

A light rain arrived overnight. The wall beneath the roof edge stayed dry, but most of the grid faded into pale smudges. Tariq suggested beginning the mural anyway and judging the proportions by eye. Amaya looked at the wide blank wall, then at the detailed sketch. A small error in one corner, she realized, could grow into a crooked wing several feet away.

Three chalk marks had survived beneath the roof edge. Amaya measured the distance between them and compared it with the sketch. She and Tariq tied a string to the first mark, checked it with a level, and snapped fresh horizontal lines across the wall. Then they measured downward to rebuild the vertical lines.

The work delayed painting by an hour, but the new grid matched the original plan. At the unveiling, visitors noticed the bright birds, not the lines hidden beneath them. Amaya did. To her, the invisible grid was evidence that careful preparation could be rebuilt, even after it seemed to disappear.

STIMULUS END

TACHS-R3-014

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S04

Prompt:

Which parts of the original grid remained visible after the rain?

- A) Several vertical lines near the ground
- B) Three chalk marks beneath the roof edge
- C) The complete outline of one bird
- D) A row of squares beside the doorway

TACHS-R3-015

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S04

Prompt:

Why does Amaya decide not to judge the mural proportions by eye?

- A) She believes the wall is too wet to paint.
- B) She wants visitors to see the chalk lines.
- C) She has forgotten which colors belong in the design.
- D) She understands that a small measuring error could distort a large part of the mural.

TACHS-R3-016

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R3-S04

Prompt:

Which lesson is best supported by Amaya's response to the erased grid?

- A) Careful methods can restore progress after an unexpected setback.
- B) Outdoor artwork should be completed only during dry seasons.
- C) A successful mural depends more on color than on measurement.
- D) Visitors appreciate artwork more when they understand every preparation step.

TACHS-R3-017

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S04

Prompt:

As used in the third paragraph, "tied a string to the first mark" most nearly shows that the string was

- A) measured loosely.
- B) hidden from view.
- C) secured at one point.
- D) colored with chalk.

STIMULUS START

Stimulus ID: TACHS-R3-S05

Section: Reading

Stimulus Type: Scientific or Technical Passage

Title: How Gecko Feet Hold On

Text:

A gecko can race across a wall and pause on a ceiling without using glue, claws, or suction cups. The secret lies in structures too small to see without magnification. Each toe is covered with millions of microscopic hairs. Every hair branches into even finer tips that can make extremely close contact with a surface.

When those tips approach a wall, weak attractions form between molecules in the toe and molecules in the wall. One attraction is tiny, but millions acting together can support the gecko's weight. Because the hairs are flexible, they conform to small bumps and dips that would prevent a stiff surface from touching evenly.

Attachment alone would not be useful if the foot could not release. A gecko changes the angle of its toes as it steps. This motion peels the hair tips away in sequence, reducing their contact with the surface. The foot can therefore grip strongly in one position and release quickly in another.

Researchers study gecko feet while developing dry adhesives for robots and reusable fasteners. A successful material must do more than stick. It must also release on command, work repeatedly, and avoid leaving residue. The gecko's foot demonstrates how many weak contacts, arranged and controlled carefully, can produce a strong but reversible grip.

STIMULUS END

TACHS-R3-018

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S05

Prompt:

What covers each gecko toe according to the passage?

- A) Millions of microscopic hairs
- B) A thin layer of natural glue
- C) Several flexible suction cups

D) Rows of curved claws

TACHS-R3-019

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S05

Prompt:

Why does the author explain how a gecko changes the angle of its toes?

- A) To show why geckos move more slowly on ceilings
- B) To compare gecko toes with the claws of other animals
- C) To show that the grip can be released as well as created
- D) To argue that toe angle matters more than microscopic hairs

TACHS-R3-020

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R3-S05

Prompt:

How is the passage mainly organized?

- A) It lists several animals and ranks their climbing ability.
- B) It presents competing theories and proves that one is false.
- C) It describes an invention and then traces its commercial history.
- D) It explains a natural mechanism, describes how it releases, and connects it to possible designs.

TACHS-R3-021

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S05

Prompt:

As used in the second paragraph, "conform" most nearly means

- A) become harder over time.
- B) fit closely to the shape of.
- C) move in a straight direction.
- D) produce a chemical coating.

STIMULUS START

Stimulus ID: TACHS-R3-S06

Section: Reading

Stimulus Type: Informational or Historical-Style Passage

Title: When Screw Threads Became Predictable

Text:

A screw may look simple, but its spiral ridge has several important measurements. The diameter, the distance between ridges, and the shape of each ridge determine which nut will fit. In early workshops, craftspeople often cut threads with tools made for their own shop. A bolt produced in one place might not match a nut made elsewhere, even when both appeared to be the same size.

As machines became more widely used, this mismatch created practical problems. A damaged bolt could delay a repair while a replacement was filed or recut by hand. Merchants also had to store many nearly identical parts because customers used different thread patterns.

Manufacturers gradually adopted shared sets of thread sizes. Gauges allowed workers to check whether a new bolt matched an agreed diameter and spacing. Printed tables helped workshops select the same measurements, and suppliers could label parts more consistently.

The change required effort. Some shops replaced cutting tools, updated records, and trained workers to use the new gauges. Standardization did not make every screw identical; different jobs still required different sizes and strengths. It created recognized families of parts. Once those

families were widely understood, repairs became more predictable, inventories became simpler, and parts made in distant workshops were more likely to fit together.

STIMULUS END

TACHS-R3-022

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S06

Prompt:

What did workers use to check whether a bolt matched an agreed thread size?

- A) Polishing stones
- B) Shipping labels
- C) Wooden patterns
- D) Gauges

TACHS-R3-023

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S06

Prompt:

What is the author's main purpose in the passage?

- A) To persuade readers that all machines should use one screw size
- B) To explain why shared thread standards developed and what problems they reduced
- C) To describe how one craftsperson invented the first threaded bolt
- D) To compare the strength of metal and wooden fasteners

TACHS-R3-024

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R3-S06

Prompt:

Which statement best summarizes the passage?

- A) Thread gauges eliminated the need for skilled workers in machine shops.
- B) Merchants preferred storing many similar parts because customers requested variety.
- C) Shared thread measurements made replacement parts and repairs more dependable across workshops.
- D) Standardization required every screw to have the same diameter and strength.

TACHS-R3-025

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S06

Prompt:

Which inference is best supported about workshops during the transition to shared standards?

- A) Some had to change tools and records before receiving the full benefits.
- B) Most stopped producing threaded parts until every shop agreed.
- C) They no longer needed to label replacement parts.
- D) They used fewer measurements because gauges replaced written tables.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE3-S03

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Textile Collection Day

Text:

[1] Our environmental club organized a textile collection day for clothing and household fabric that families no longer used. [2] The goal was to keep usable cloth in circulation and damaged fabric out of the trash. [3] We placed labeled bins for wearable items, torn cotton, and mixed fabric beside the gym entrance. [4] Volunteers checked each bag before placing its contents in the correct bin. [5] Volunteers weighed each group before packing it. [6] A local shelter accepted the wearable items, while a recycling company took fabric that could not be reused. [7] By the end of the day, the bins held 186 pounds of material. [8] The project showed that clear sorting instructions can turn unwanted fabric into useful resources.

STIMULUS END

TACHS-WE3-014

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S03

Prompt:

Which sentence would best follow sentence 2 to develop the explanation of the collection plan?

- A) The club had held several meetings in the same gym during the year.
- B) Some volunteers wore blue shirts, while others wore green shirts.
- C) To support that goal, the club separated wearable items from fabric that required recycling.
- D) The collection day began on a Saturday morning in early spring.

TACHS-WE3-015

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S03

Prompt:

Which revision of sentence 5 is most precise?

- A) Volunteers weighed the fabric in each category before packing it into labeled boxes.
- B) Volunteers did something with every group before putting it away.
- C) Before packing, the groups were weighed by volunteers in a way.
- D) Volunteers weighed them before they packed all of it.

TACHS-WE3-016

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S03

Prompt:

Which revision most effectively combines sentences 1 and 2?

- A) Our environmental club organized a textile collection day, families no longer used clothing and household fabric.
- B) The goal was to keep usable cloth in circulation, our environmental club organized a collection day.
- C) Our environmental club organized a textile collection day and the goal, keeping fabric out of the trash.
- D) To keep usable cloth in circulation and damaged fabric out of the trash, our environmental club organized a textile collection day.

TACHS-WE3-017

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S03

Prompt:

Which sentence would best replace sentence 8 as a conclusion?

- A) Some families brought fabric in paper bags, while others used boxes.
- B) Because the bins were clearly labeled, families sorted materials quickly and helped redirect 186 pounds of fabric.

- C) The club may choose a different gym entrance for its next event.
- D) Wearable clothing is available in many colors and sizes.

STIMULUS START

Stimulus ID: TACHS-WE3-S04

Section: Written Expression

Stimulus Type: Revision Passage

Title: Testing Parcel Padding

Text:

[1] Our engineering class compared three recycled materials for protecting a fragile object inside a package. [2] Each team placed the same type of clay cup in an identical box. [3] Using shredded paper, folded cardboard, or crumpled cloth, the boxes were packed. [4] We dropped every box from a height of one meter onto the same rubber mat. [5] Before each trial, the cups were checked for cracks. [6] The cloth-packed boxes protected all five cups, while the other materials protected four cups each. [7] The classroom windows faced east. [8] We concluded that cloth performed best in our small test, but more trials would be needed before making a broad claim.

STIMULUS END

TACHS-WE3-018

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S04

Prompt:

Where should the following sentence be placed? Each padding material was tested in five separate boxes.

- A) Before sentence 1
- B) After sentence 3
- C) After sentence 6
- D) After sentence 8

TACHS-WE3-019

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S04

Prompt:

Which revision best corrects the misplaced modifier in sentence 3?

- A) Using shredded paper, folded cardboard, or crumpled cloth, identical boxes were available.
- B) The boxes, using shredded paper, folded cardboard, or crumpled cloth, were packed by materials.
- C) The teams packed the boxes using shredded paper, folded cardboard, or crumpled cloth.
- D) Shredded paper, folded cardboard, or crumpled cloth, the teams packed the boxes.

TACHS-WE3-020

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S04

Prompt:

Which revision of sentence 6 states the result most precisely?

- A) All five cups packed with cloth remained uncracked, compared with four of five cups in each other group.
- B) The cloth worked better, while the other materials were almost as good somehow.
- C) Most cups survived, and cloth was a material used in some boxes.
- D) The boxes with cloth had cups, but the other boxes also had cups that were protected.

TACHS-WE3-021

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE3-S04

Prompt:

Which revision of sentence 5 uses consistent past tense and a clear subject?

- A) Before each trial, the cups are checked for cracks.
- B) Before each trial, checking the cups for cracks.
- C) Before each trial, the cups had checked for cracks.
- D) Before each trial, we checked each cup for cracks.

TACHS-WE3-022

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) A box of old cables were stored beneath the table.
- B) A box of old cables have been stored beneath the table.
- C) A box of old cables was stored beneath the table.
- D) A box of old cables store beneath the table.

TACHS-WE3-023

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses pronouns correctly?

- A) The teacher gave Luis and I the updated schedule.
- B) The teacher gave Luis and me the updated schedule.
- C) The teacher gave he and Luis the updated schedule.
- D) The teacher gave Luis and myself the updated schedule.

TACHS-WE3-024

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The volunteers carried the labeled bins outside.

Line 2: Each bin was placed beside the correct truck.

Line 3: The coordinator checked the final count.

Line 4: Neither of the labels were attached securely.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE3-025

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The donations arrived before noon.

Line 2: Two students recorded the weight of each bag.

Line 3: The volunteers sorted the materials careful.

Line 4: Later, the team cleaned the collection area.

- A) Line 1

- B) Line 2
- C) Line 3
- D) Line 4

TACHS-WE3-026

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: Each of the posters have a date printed at the bottom.

Line 2: The largest poster hangs beside the entrance.

Line 3: Visitors can read the schedule from the hallway.

Line 4: A smaller copy is posted near the stairs.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M3-013

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $2\frac{1}{3} \div \frac{7}{9}$?

- A) $1\frac{5}{9}$
- B) $2\frac{1}{9}$
- C) 3
- D) $4\frac{2}{3}$

TACHS-M3-014

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A supply room has 84 pencils and 126 erasers. What is the greatest number of identical kits that can be made using all the items?

- A) 21
- B) 42
- C) 84
- D) 126

TACHS-M3-015

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which expression has a value of -14 ?

- A) $-3 \times 6 + 4$
- B) $18 \div (-3) + 4$
- C) $-20 - (-8)$
- D) $5 - 3 \times 4$

TACHS-M3-016

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Which inequality is the solution to $3x + 5 \leq 20$?

A) $x \geq 5$

B) $x < 5$

C) $x > 5$

D) $x \leq 5$

TACHS-M3-017

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A function has the values $x = 1, 3, 5$ and $y = 4, 10, 16$. Which rule describes the function?

A) $y = 2x + 2$

B) $y = 3x + 1$

C) $y = 3x - 1$

D) $y = 4x$

TACHS-M3-018

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The six measurements are 14, 18, 19, 21, 26, and 32. What is the median?

A) 19

B) 19.5

C) 20

D) 21

TACHS-M3-019

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A spinner has eight equal sectors labeled A, A, A, B, B, C, D, and D. What is the probability of landing on A?

A) $\frac{3}{8}$

B) $\frac{1}{2}$

C) $\frac{5}{8}$

D) $\frac{3}{5}$

TACHS-M3-020

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

Of 120 students surveyed, 42 ride a bus, 30 walk, and 18 ride a bicycle. The rest travel by car. What percent travel by car?

A) 20%

B) 22.5%

C) 24%

D) 25%

TACHS-M3-021

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

What is the distance between the points $(-3, 4)$ and $(5, 4)$?

- A) 2 units
- B) 6 units
- C) 8 units
- D) 12 units

TACHS-M3-022

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Two triangles are similar. A side of 6 centimeters on the smaller triangle corresponds to a side of 10 centimeters on the larger triangle. If another side of the smaller triangle is 9 centimeters, what is the corresponding side of the larger triangle?

- A) 12 centimeters
- B) 13.5 centimeters
- C) 14 centimeters
- D) 15 centimeters

TACHS-M3-023

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A trapezoid has parallel bases of 8 centimeters and 14 centimeters and a height of 5 centimeters. What is its area?

- A) 44 square centimeters
- B) 55 square centimeters
- C) 70 square centimeters
- D) 110 square centimeters

TACHS-M3-024

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A roll contains 4 yards 2 feet of ribbon. How many feet of ribbon are on the roll?

- A) 14 feet
- B) 12 feet
- C) 10 feet
- D) 6 feet

TACHS-M3-025

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A package has a mass of 2.4 kilograms. An additional item has a mass of 650 grams. What is the combined mass in kilograms?

- A) 2.465 kilograms
- B) 2.95 kilograms
- C) 3.00 kilograms
- D) 3.05 kilograms

SECTION 4 - ABILITY

TACHS-A3-013

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×3 matrix.

Row 1: ○○○○ | ○ | ●●●

Row 2: △△△△ | △△ | ?

In each row, subtract the second count from the first count, then change the remaining open shapes to filled shapes.

- A) △△△
- B) ▲▲
- C) △△
- D) ▲▲▲

TACHS-A3-014

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: Top ○; Bottom □

Top-right: Left ○; Right □

Bottom-left: Top △; Bottom ◇

Bottom-right: ?

Across each row, the vertical arrangement turns 90° counterclockwise without changing symbol order.

- A) Left ◇; Right △
- B) Left △; Right ◇
- C) Top ◇; Bottom △
- D) Top △; Bottom ◇

TACHS-A3-015

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ○ | ●● | ○○○

Row 2: □□ | ■■■ | □

Row 3: △△△ | ▲ | ?

The counts 1, 2, and 3 shift one position left in each new row. The center cell in every row is filled.

- A) △△
- B) ▲▲
- C) △
- D) ▲▲▲

TACHS-A3-016

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: 

Top-right: 

Bottom-left: 

Bottom-right: ?

Moving right fills only the first shape. Moving down fills only the second shape.

A) 

B) 

C) 

D) 

TACHS-A3-017

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular sheet is divided into three equal vertical panels. The left panel is folded right onto the center panel, and the right panel is folded left onto the same center panel. One hole is punched near the top of the final stack, away from every edge and fold. What appears after unfolding?

A) Two holes in a horizontal row

B) Three holes in a horizontal row, one in each panel

C) Three holes in a vertical column

D) Six holes arranged in two rows

TACHS-A3-018

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded once from left to right. Two holes are punched in the folded rectangle: one near the upper-right corner and one near the lower-right corner, both away from every edge and the fold. What appears after unfolding?

A) Two holes near the right edge

B) Two holes along the vertical centerline

C) Three holes forming a triangle

D) Four holes, one near each corner

TACHS-A3-019

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular sheet is folded from top to bottom. The folded rectangle is then divided into three equal vertical panels; the left panel is folded onto the center, and the right panel is folded onto the center. One interior hole is punched through the final stack. What appears after unfolding?

A) Three holes in one horizontal row

B) Four holes in a 2×2 arrangement

C) Six holes in a 2×3 arrangement

D) Eight holes in two rows

TACHS-A3-020

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular sheet is folded in half from left to right. A semicircular notch is cut at the midpoint of the folded packet's right outer edge, away from the fold. What appears after unfolding?

- A) Two matching semicircular notches, one at the midpoint of each outer vertical edge
- B) One circular hole at the center of the sheet
- C) Two notches along the top edge
- D) Four semicircular notches, one near each corner

TACHS-A3-021

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

○ □ ■ ●

△ ◇ ◆ ▲

☆ ○ ● ★

Which choice follows the same rule?

A) □ △ ▲ ■

B) □ △ ■ ▲

C) ■ ▲ △ □

D) □ ▲ △ ■

TACHS-A3-022

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

○ ■■

△ ◆◆

☆ ●●

Which choice follows the same rule?

A) □ □□

B) ■ △△

C) □ ▲▲

D) ◆ ○○

TACHS-A3-023

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share the same rule:

↑ ○ ●

→ □ ■

↓ △ ▲

Which choice follows the same rule?

A) ← ◆ ◇

B) ← ◇ □

C) ← ■ □

D) ← ◇ ◆

TACHS-A3-024

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share the same rule:

| - /

/ | -

- / |

Which choice follows the same rule?

A) | | /

B) | / -

C) - - |

D) / \ /

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R3-014 - Correct: B - Correct Answer: Three chalk marks beneath the roof edge - Explanation: The passage states that three chalk marks survived in the dry area beneath the roof edge.

TACHS-R3-015 - Correct: D - Correct Answer: She understands that a small measuring error could distort a large part of the mural. - Explanation: Amaya realizes that a minor error on the sketch scale could become a large distortion on the wall.

TACHS-R3-016 - Correct: A - Correct Answer: Careful methods can restore progress after an unexpected setback. - Explanation: Amaya and Tariq recover from the erased grid by using surviving evidence and careful measurement.

TACHS-R3-017 - Correct: C - Correct Answer: secured at one point. - Explanation: The string is fixed to the first mark so it can guide a straight replacement line.

TACHS-R3-018 - Correct: A - Correct Answer: Millions of microscopic hairs - Explanation: The first paragraph identifies millions of microscopic hairs as the structures covering each toe.

TACHS-R3-019 - Correct: C - Correct Answer: To show that the grip can be released as well as created - Explanation: Changing toe angle peels the hair tips away, showing how the gecko controls release after gripping.

TACHS-R3-020 - Correct: D - Correct Answer: It explains a natural mechanism, describes how it releases, and connects it to possible designs. - Explanation: The passage moves from attachment to controlled release and then to engineering applications.

TACHS-R3-021 - Correct: B - Correct Answer: fit closely to the shape of. - Explanation: Flexible hairs conform by fitting closely around tiny bumps and dips in a surface.

TACHS-R3-022 - Correct: D - Correct Answer: Gauges - Explanation: The passage states that gauges checked whether bolts matched agreed diameters and spacing.

TACHS-R3-023 - Correct: B - Correct Answer: To explain why shared thread standards developed and what problems they reduced - Explanation: The passage explains the mismatch problem, the adoption process, and the practical benefits of standards.

TACHS-R3-024 - Correct: C - Correct Answer: Shared thread measurements made replacement parts and repairs more dependable across workshops. - Explanation: Recognized thread families made parts more interchangeable without requiring every screw to be identical.

TACHS-R3-025 - Correct: A - Correct Answer: Some had to change tools and records before receiving the full benefits. - Explanation: The passage notes that shops replaced tools, updated records, and trained workers during adoption.

WRITTEN EXPRESSION ANSWERS

TACHS-WE3-014 - Correct: C - Correct Answer: To support that goal, the club separated wearable items from fabric that required recycling. - Explanation: The added sentence directly connects the stated goal with the sorting system described next.

TACHS-WE3-015 - Correct: A - Correct Answer: Volunteers weighed the fabric in each category before packing it into labeled boxes. - Explanation: The revision identifies what was weighed and how the sorted fabric was packed.

TACHS-WE3-016 - Correct: D - Correct Answer: To keep usable cloth in circulation and damaged fabric out of the trash, our environmental club organized a textile collection day. - Explanation: The introductory purpose phrase clearly and correctly connects the goal with the club's action.

TACHS-WE3-017 - Correct: B - Correct Answer: Because the bins were clearly labeled, families sorted materials quickly and helped redirect 186 pounds of fabric. - Explanation: The revision connects the sorting method with the project's measured result and central purpose.

TACHS-WE3-018 - Correct: B - Correct Answer: After sentence 3 - Explanation: The sentence belongs after the materials are introduced and before the drop procedure and results.

TACHS-WE3-019 - Correct: C - Correct Answer: The teams packed the boxes using shredded paper, folded cardboard, or crumpled cloth. - Explanation: The revision makes "the teams" the clear performers of the action described by the opening phrase.

TACHS-WE3-020 - Correct: A - Correct Answer: All five cups packed with cloth remained uncracked, compared with four of five cups in each other group. - Explanation: The revision gives exact results for the cloth group and both comparison groups.

TACHS-WE3-021 - Correct: D - Correct Answer: Before each trial, we checked each cup for cracks. - Explanation: "We checked" provides a clear subject and matches the passage's past-tense narration.

TACHS-WE3-022 - Correct: C - Correct Answer: A box of old cables was stored beneath the table. - Explanation: The singular subject "box" requires the singular verb "was."

TACHS-WE3-023 - Correct: B - Correct Answer: The teacher gave Luis and me the updated schedule. - Explanation: The object of "gave" requires the objective pronoun "me."

TACHS-WE3-024 - Correct: D - Correct Answer: Line 4 - Explanation: The singular subject "neither" requires "was," not "were."

TACHS-WE3-025 - Correct: C - Correct Answer: Line 3 - Explanation: The adverb "carefully," not the adjective "careful," must modify "sorted."

TACHS-WE3-026 - Correct: A - Correct Answer: Line 1 - Explanation: The singular subject "each" requires "has," not "have."

MATHEMATICS ANSWERS

TACHS-M3-013 - Correct: C - Correct Answer: 3 - Explanation: Convert $2\frac{1}{3}$ to $\frac{7}{3}$, then multiply by $\frac{9}{7}$: $\frac{7}{3} \times \frac{9}{7} = 3$.

TACHS-M3-014 - Correct: B - Correct Answer: 42 - Explanation: The greatest common factor of 84 and 126 is 42, so 42 identical kits can be made.

TACHS-M3-015 - Correct: A - Correct Answer: $-3 \times 6 + 4$ - Explanation: Apply order of operations: $-3 \times 6 + 4 = -18 + 4 = -14$.

TACHS-M3-016 - Correct: D - Correct Answer: $x \leq 5$ - Explanation: Subtract 5 and divide by 3: $3x \leq 15$, so $x \leq 5$.

TACHS-M3-017 - Correct: B - Correct Answer: $y = 3x + 1$ - Explanation: Each y-value is one more than three times x, so the rule is $y = 3x + 1$.

TACHS-M3-018 - Correct: C - Correct Answer: 20 - Explanation: The middle values are 19 and 21, and their average is 20.

TACHS-M3-019 - Correct: A - Correct Answer: $\frac{3}{8}$ - Explanation: Three of the eight equal sectors are labeled A, so the probability is $\frac{3}{8}$.

TACHS-M3-020 - Correct: D - Correct Answer: 25% - Explanation: The car group has $120 - 42 - 30 - 18 = 30$ students, and $30/120 = 25\%$.

TACHS-M3-021 - Correct: C - Correct Answer: 8 units - Explanation: The points share the same y-coordinate, and $5 - (-3) = 8$ units.

TACHS-M3-022 - Correct: D - Correct Answer: 15 centimeters - Explanation: The scale factor is $\frac{10}{6}$, so the corresponding side is $9 \times \frac{10}{6} = 15$ centimeters.

TACHS-M3-023 - Correct: B - Correct Answer: 55 square centimeters - Explanation: Area is $\frac{1}{2} \times (8 + 14) \times 5 = 55$ square centimeters.

TACHS-M3-024 - Correct: A - Correct Answer: 14 feet - Explanation: Four yards equal 12 feet, and $12 + 2 = 14$ feet.

TACHS-M3-025 - Correct: D - Correct Answer: 3.05 kilograms - Explanation: Convert 650 grams to 0.65 kilogram, then add $2.4 + 0.65 = 3.05$ kilograms.

ABILITY ANSWERS

TACHS-A3-013 - Correct: D - Correct Answer: ▲▲ - Explanation: Five triangles minus two triangles leaves three, and the rule changes the remaining shapes to filled triangles.

TACHS-A3-014 - Correct: B - Correct Answer: Left $\triangle$; Right $\diamond$ - Explanation: Turning the vertical pair counterclockwise places the top triangle on the left and the bottom diamond on the right.

TACHS-A3-015 - Correct: A - Correct Answer: $\triangle\triangle$ - Explanation: The third row must continue the shifted count pattern 3, 1, 2, and the last cell is open.

TACHS-A3-016 - Correct: C - Correct Answer: $\bullet\blacksquare$ - Explanation: Applying both independent changes fills the circle and the square, producing $\bullet\blacksquare$.

TACHS-A3-017 - Correct: B - Correct Answer: Three holes in a horizontal row, one in each panel - Explanation: The three stacked vertical panels each receive the punch, producing one top-position hole in every panel.

TACHS-A3-018 - Correct: D - Correct Answer: Four holes, one near each corner - Explanation: Each punch reflects across the vertical fold, producing matching upper and lower pairs near all four corners.

TACHS-A3-019 - Correct: C - Correct Answer: Six holes in a 2×3 arrangement - Explanation: The horizontal fold doubles the three vertical layers, producing six reflected holes in two rows of three.

TACHS-A3-020 - Correct: A - Correct Answer: Two matching semicircular notches, one at the midpoint of each outer vertical edge - Explanation: The stacked original outer edges are cut together, so unfolding places matching notches on the left and right edges.

TACHS-A3-021 - Correct: A - Correct Answer: $\square \triangle \blacktriangle \blacksquare$ - Explanation: Each arrangement shows two open shapes followed by their filled counterparts in reverse order.

TACHS-A3-022 - Correct: C - Correct Answer: $\square \blacktriangle \blacktriangle$ - Explanation: Each arrangement begins with one open shape and ends with two filled copies of a different shape.

TACHS-A3-023 - Correct: D - Correct Answer: $\leftarrow \diamond \blacklozenge$ - Explanation: Each arrangement has one arrow followed by an open shape and then that shape's filled counterpart.

TACHS-A3-024 - Correct: B - Correct Answer: $| / -$ - Explanation: Each valid group contains exactly one vertical, one horizontal, and one forward-diagonal line.

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R3-S07

Section: Reading

Stimulus Type: Literary Passage

Title: The Last Lantern

Text:

Mina had promised to bring a lantern to the neighborhood's evening garden walk. Her grandfather's brass lantern seemed perfect until she discovered that its glass door would not close. With an hour remaining, she carried it to Mr. Bell's repair shop.

Mr. Bell examined the bent latch. "I can straighten this," he said, "but brass remembers every impatient hand." Instead of reaching for a heavy tool, he warmed the latch beneath a small lamp and pressed it gradually with padded pliers. Mina watched the metal move by fractions.

"Wouldn't hitting it be faster?" she asked.

"For a moment," he replied. "Then we might spend tomorrow repairing what speed broke today."

When the latch finally clicked, Mina polished the cloudy glass. At the walk, she carried the lantern at the rear of the group. Its modest circle of light did not illuminate the entire path, but it revealed each stone before someone stepped on it. Mina stopped wishing she had brought the brightest lantern. She began matching her pace to the people ahead, keeping the light where it was needed.

The next morning, she returned the lantern to its shelf. Beside it, she placed a note: "Handle slowly. Useful things do not have to hurry."

STIMULUS END

TACHS-R3-026

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S07

Prompt:

Why does Mina visit Mr. Bell?

- A) The lantern's latch is bent.
- B) The lantern has run out of fuel.
- C) She wants to buy padded pliers.
- D) She needs directions to the garden.

TACHS-R3-027

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S07

Prompt:

What does Mr. Bell most likely mean when he says that “brass remembers every impatient hand”?

- A) Old objects can identify their owners.
- B) Brass becomes brighter when handled.
- C) Careless force can leave lasting damage.
- D) Repairs should always be completed by hand.

TACHS-R3-028

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S07

Prompt:

As used in the passage, “modest” most nearly means

- A) newly repaired.
- B) limited but adequate.
- C) difficult to notice.
- D) unusually beautiful.

TACHS-R3-029

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R3-S07

Prompt:

Which statement best expresses a central idea of the passage?

- A) Antique objects should be displayed rather than used.
- B) The brightest tool is generally the most dependable.
- C) Community events require careful advance planning.
- D) Patience can protect useful things and help people serve others well.

STIMULUS START

Stimulus ID: TACHS-R3-S08

Section: Reading

Stimulus Type: Literary Passage

Title: A Different Route

Text:

The trail map showed one straight path to the overlook, and Eli had memorized every turn. Halfway up the hill, however, a fallen oak blocked the narrow trail. Its branches stretched into thorny brush on both sides.

Eli's cousin Nia pointed to a faint path descending toward the creek. "The maintenance crew may use that route," she said. Eli disliked walking downhill when their destination was above them, but climbing over the unstable trunk was clearly unsafe.

They followed the faint path. Near the creek, the trail disappeared beneath smooth stones. Nia noticed three small blue marks painted on rocks at regular intervals. The marks led upstream, crossed a wooden footbridge, and joined the main trail beyond the fallen tree.

At the overlook, Eli unfolded his map. The maintenance route was absent. He sketched it lightly in the margin, including the bridge and blue markers, but added a question mark beside the entrance. On the return trip, they met a ranger, who confirmed that the route was open temporarily while crews cleared the oak.

Eli erased the question mark and wrote "temporary detour." He had always treated maps as finished answers. Now he understood them as records that sometimes needed careful updates from the ground beneath his feet.

STIMULUS END

TACHS-R3-030

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S08

Prompt:

What helps Eli and Nia follow the route near the creek?

- A) Blue marks painted on rocks
- B) Arrows carved into trees
- C) Directions printed on the map
- D) Signs posted by the ranger

TACHS-R3-031

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S08

Prompt:

Why does Eli first place a question mark beside the route he sketches?

- A) He cannot remember where the route begins.
- B) He is uncertain whether the route is approved for hikers.
- C) He plans to ask Nia to redraw the bridge.
- D) He believes the blue marks indicate danger.

TACHS-R3-032

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R3-S08

Prompt:

How does the author mainly develop Eli's changing view of maps?

- A) By comparing maps produced in several periods
- B) By describing the ranger's daily responsibilities
- C) By presenting a problem that experience helps Eli reconsider
- D) By explaining every step used to print a trail map

TACHS-R3-033

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S08

Prompt:

Which word best describes the author's tone?

- A) Playful
- B) Alarmed
- C) Disapproving
- D) Thoughtful

STIMULUS START

Stimulus ID: TACHS-R3-S09

Section: Reading

Stimulus Type: Informational Passage

Title: Why Some Cities Build Rain Gardens

Text:

During a storm, rain falling on roofs, sidewalks, and parking lots cannot soak directly into soil. Instead, it flows across these hard surfaces, gathering leaves, dirt, and other materials before entering street drains. A large volume arriving quickly can burden drainage systems and carry pollutants into streams.

A rain garden is a shallow planted basin designed to receive some of this runoff. Water enters the basin, spreads among grasses and flowering plants, and slowly filters through layers of soil. The garden is not a permanent pond. Designers select its depth, soil mixture, and drainage area so that standing water usually disappears within a day or two.

Plant choice matters. Deep-rooted native species can loosen soil and tolerate alternating wet and dry periods. Their stems also slow moving water. Yet a rain garden must be placed carefully. It should not sit over underground utilities or directly against a building foundation, and its size must match the area draining into it.

Rain gardens cannot prevent every flood or remove every pollutant. Their value comes from handling many smaller flows before those flows combine. When numerous properties use them, the gardens function like a scattered network of small sponges, reducing pressure on a city's drainage system while creating habitat for insects and birds.

STIMULUS END

TACHS-R3-034

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S09

Prompt:

According to the passage, why are deep-rooted native plants useful in rain gardens?

- A) They loosen soil and tolerate changing moisture levels.
- B) They prevent all water from reaching street drains.
- C) They require the basin to remain permanently flooded.
- D) They protect underground utilities from damage.

TACHS-R3-035

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S09

Prompt:

As used in the first paragraph, “burden” most nearly means

- A) repair.
- B) conceal.
- C) place excessive demand on.
- D) carry beneath the ground.

TACHS-R3-036

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R3-S09

Prompt:

Which statement best summarizes the passage?

- A) Rain gardens replace city drains by holding storm water permanently.
- B) Rain gardens use carefully chosen sites, soils, and plants to slow and filter runoff, though they have limits.
- C) Native plants survive only when they grow near building foundations.
- D) Most storm pollution comes from soil inside planted basins.

TACHS-R3-037

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S09

Prompt:

Why does the author compare many rain gardens to “a scattered network of small sponges”?

- A) To show that each garden must be made from an artificial material
- B) To suggest that rain gardens make city soil permanently wet
- C) To explain why insects build homes beneath the plants
- D) To emphasize their combined ability to absorb and slow water in many locations

TACHS-R3-038

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S09

Prompt:

What is the author's main purpose?

- A) To explain how rain gardens work and identify both their benefits and limitations
- B) To persuade readers that every lawn should be replaced by a pond
- C) To criticize cities that maintain underground drainage systems
- D) To provide instructions for identifying every native plant species

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE3-S05

Section: Written Expression

Stimulus Type: Revision Passage

Title: Recording Local Sounds

Text:

[1] Our media club decided to create an audio map of the neighborhood. [2] Teams recorded short sounds at the market, train platform, park, and riverside path. [3] We used microphones to make recordings of sounds that could be heard. [4] Back at school, students labeled each clip with its location and time. [5] Because wind covered several voices, those interviews had to be recorded again. [6] Next, the club arranged the clips to guide listeners from the busiest place to the quietest. [7] The cafeteria serves vegetable soup on Wednesdays. [8] A brief spoken introduction explains why familiar places can sound different at different hours. [9] The completed map will be posted on the school website.

STIMULUS END

TACHS-WE3-027

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S05

Prompt:

Which revision of sentence 3 is most concise?

- A) We made recordings with microphones of sounds that people were able to hear.
- B) Microphones were the equipment used by us for audible recordings.
- C) We recorded the sounds with microphones.

D) Using microphones, sounds that could be heard were recorded by our club.

TACHS-WE3-028

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S05

Prompt:

Which sentence should be removed because it does not support the passage's central idea?

A) Sentence 7

B) Sentence 4

C) Sentence 6

D) Sentence 8

TACHS-WE3-029

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S05

Prompt:

Which revision most effectively combines sentences 4 and 5?

A) Back at school, students labeled each clip, and its location and time, wind covered voices.

B) After students labeled each clip with its location and time, they rerecorded interviews in which wind had covered voices.

C) Students labeled clips with locations and times, because recording the interviews again.

D) Labeling every clip back at school, several voices covered by wind and repeated.

TACHS-WE3-030

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S05

Prompt:

Which phrase best replaces “the busiest place to the quietest” in sentence 6 to make the comparison precise?

A) one place and then another place

B) locations with a whole lot of noise toward calmer ones

C) the places that were busy through those not busy

D) the location with the loudest activity to the location with the least

STIMULUS START

Stimulus ID: TACHS-WE3-S06

Section: Written Expression

Stimulus Type: Revision Passage

Title: The Seed Library

Text:

[1] A seed library allows gardeners to borrow packets of seeds and later donate seeds they collect. [2] Our community center opened one in a wooden cabinet near the entrance. [3] Volunteers sorted packets by plant type, they also added labels showing planting depth and season. [4] Visitors can choose seeds without paying, but they are encouraged to return some after a successful harvest. [5] This exchange preserves locally successful varieties. [6] It also gives beginning gardeners an inexpensive way to experiment. [7] Clear instructions beside the cabinet explain how to dry and label donated seeds. [8] The seed library is small, yet each returned envelope helps the collection continue.

STIMULUS END

TACHS-WE3-031

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S06

Prompt:

Which revision correctly repairs sentence 3?

- A) Volunteers sorted packets by plant type and adding labels that showed planting depth and season.
- B) Volunteers sorted packets by plant type; they also added labels showing planting depth and season.
- C) Volunteers sorted packets by plant type, also labels showing planting depth and season.
- D) Volunteers sorting packets by plant type while labels showed planting depth and season.

TACHS-WE3-032

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S06

Prompt:

Where should the following sentence be placed?

Some packets contain beans and herbs that families have grown in the area for decades.

- A) After sentence 5
- B) After sentence 2
- C) Before sentence 1
- D) After sentence 8

TACHS-WE3-033

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S06

Prompt:

Which revision of sentence 8 is most effective?

- A) Though modest in size, the seed library grows stronger with every properly returned envelope.
- B) The seed library is little, but envelopes are things that make it go.
- C) Small is the seed library, and returning is continuing it.
- D) The cabinet has envelopes, yet it is a small kind of library.

TACHS-WE3-034

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) Each of the painted signs have a protective coating.
- B) Neither the coach nor the players was expecting rain.
- C) The collection of old postcards belong in this drawer.
- D) Every one of the volunteers receives a name tag.

TACHS-WE3-035

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses the pronoun correctly?

- A) The librarian gave Luis and I the cabinet key.
- B) Between you and me, the final clue was difficult.
- C) Her and Malik finished the display before noon.
- D) The judges congratulated she after the contest.

TACHS-WE3-036

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: None

Prompt:

Which sentence has correct parallel structure?

- A) The internship required arriving early, careful notes, and to ask questions.
- B) Our goals were to reduce waste, saving money, and cleaner rooms.
- C) The guide taught us how to measure, record, and compare the samples.
- D) Nina enjoys sketching buildings, to photograph doors, and maps.

TACHS-WE3-037

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: The museum guide spoke clearly.

Line 2: She described how the first machine worked.

Line 3: The visitors listened attentive to every detail.

Line 4: Afterward, they asked several questions.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

TACHS-WE3-038

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

Line 1: Before the doors opened, Marcus checked the tickets.

Line 2: He placed each stack on the correct table.

Line 3: The signs were already hanging above the entrances.

Line 4: By nine o'clock, every volunteers had arrived.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M3-026

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A jacket priced at \$64 is discounted 25%. What is the sale price?

A) \$16

B) \$39

C) \$48

D) \$52

TACHS-M3-027

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $\frac{3}{5} + \frac{7}{10} \div 2$?

A) $\frac{19}{20}$

B) $\frac{13}{20}$

C) $\frac{4}{5}$

D) $1\frac{3}{10}$

TACHS-M3-028

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A recipe uses flour and oats in the ratio 5:2. If 21 cups of the mixture are prepared, how many cups are oats?

A) 3

B) 5

C) 15

D) 6

TACHS-M3-029

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The first four terms of a sequence are 4, 9, 19, 39. Which rule produces each term after the first?

A) Add 5.

B) Multiply by 2, then add 1.

C) Multiply by 3, then subtract 3.

D) Add consecutive multiples of 5.

TACHS-M3-030

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $4(2x - 3) = 3x + 18$.

- A) $x = 3$
- B) $x = 5$
- C) $x = 6$
- D) $x = 8$

TACHS-M3-031

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A streaming service charges \$7 per month plus \$2 for each movie rented. Which expression gives the cost for m movies in one month?

- A) $7m + 2$
- B) $9m$
- C) $7(m + 2)$
- D) $7 + 2m$

TACHS-M3-032

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The numbers of pages read by five students are 18, 22, 22, 25, and 33. What is the mean?

- A) 24
- B) 22
- C) 25
- D) 120

TACHS-M3-033

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A bag contains 3 red, 2 blue, and 5 green tiles. One tile is drawn, replaced, and a second tile is drawn. What is the probability that both tiles are blue?

- A) $1/10$
- B) $1/20$
- C) $1/25$
- D) $2/25$

TACHS-M3-034

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Two angles of a triangle measure 47° and 68° . What is the measure of the third angle?

- A) 55°
- B) 65°
- C) 75°
- D) 115°

TACHS-M3-035

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangle has an area of 96 square centimeters and a length of 12 centimeters. What is its perimeter?

- A) 40 cm
- B) 32 cm
- C) 20 cm
- D) 48 cm

TACHS-M3-036

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A tank contains 3.5 liters of water. How many milliliters is this?

- A) 35
- B) 350
- C) 30,500
- D) 3,500

TACHS-M3-037

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A bus leaves at 10:47 a.m. and arrives at 1:18 p.m. How long is the trip?

- A) 2 hours 41 minutes
- B) 2 hours 31 minutes

- C) 3 hours 29 minutes
D) 3 hours 31 minutes

SECTION 4 - ABILITY

TACHS-A3-025

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix.

Row 1: ○ | ●

Row 2: □ | ?

Across each row, the shape keeps its form and changes from open to filled.

- A) ○
B) ■
C) ●
D) □

TACHS-A3-026

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix.

Row 1: ↑ | → | ↓

Row 2: → | ↓ | ←

Row 3: ↓ | ← | ?

Each step to the right or downward rotates the arrow 90° clockwise.

- A) ↑
B) →
C) ←
D) ↓

TACHS-A3-027

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix.

Row 1: ○ | ○○

Row 2: △△ | ?

Across each row, one matching shape is added.

- A) $\triangle \triangle \triangle$
- B) $\circ \circ \circ$
- C) $\triangle$
- D) $\blacktriangle \blacktriangle \blacktriangle$

TACHS-A3-028

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix.

Row 1: $\circ \square \mid \square \circ$

Row 2: $\triangle \diamond \mid ?$

Across each row, the left and right positions exchange.

- A) $\triangle \diamond$
- B) $\diamond \triangle$
- C) $\blacktriangle \blacklozenge$
- D) $\diamond \diamond$

TACHS-A3-029

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix.

Row 1: $\circ \mid \square \mid \circ \square$

Row 2: $\triangle \mid \star \mid ?$

In each row, the third cell combines the first two symbols in order.

- A) $\star \triangle$
- B) $\triangle \triangle$
- C) $\triangle \star$
- D) $\circ \square$

TACHS-A3-030

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded from top to bottom. One hole is punched near the upper-right corner of the folded rectangle, away from the fold. What appears after unfolding?

- A) Four holes, one near each corner
- B) Two holes near the upper-left and upper-right corners
- C) One hole on the horizontal centerline
- D) Two holes near the upper-right and lower-right corners

TACHS-A3-031

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded from right to left, then from top to bottom. A hole is punched near the lower-left corner of the final folded square, away from both folds. What appears after complete unfolding?

- A) Two holes reflected across only the vertical centerline
- B) Four holes, one in each quadrant, reflected across both centerlines
- C) Two holes reflected across only the horizontal centerline
- D) Eight holes arranged around the center

TACHS-A3-032

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular sheet is folded once along its vertical centerline. A triangular cut is made on the folded edge, above the midpoint. What appears after unfolding?

- A) One diamond-shaped opening above the center
- B) Two triangular notches on the outer edges
- C) Four triangular openings around the center
- D) One triangular opening on the left side

TACHS-A3-033

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded along the diagonal from upper-left to lower-right. A hole is punched near the upper-right corner of the folded triangle, away from the fold. What appears after unfolding?

- A) One hole at the center
- B) Four holes symmetric across both diagonals
- C) Two holes mirrored across the upper-left-to-lower-right diagonal
- D) Two holes on the vertical centerline

TACHS-A3-034

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three arrangements share the same rule.

1: ○ ● ○

2: □ ■ □

3: △ ▲ △

Which choice follows the same rule?

A) ○ □ ○

B) ◇ ◆ ◇

C) ◆ ◇ ◆

D) ☆ ☆ ★

TACHS-A3-035

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three arrangements share the same rule.

1: ↑ → ↓

2: → ↓ ←

3: ↓ ← ↑

Which choice follows the same rule?

A) ← ↓ ↑

B) ↑ ↓ →

C) → ↑ ←

D) ← ↑ →

TACHS-A3-036

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three figures share the same rule.

1: ○ inside □; □ inside △

2: △ inside ◇; ◇ inside ☆

3: ☆ inside ○; ○ inside □

Which choice follows the same rule?

A) ○ inside ○; ○ inside △

- B) ◆ inside □; □ outside △
 C) □ inside △; △ inside ◇
 D) △ outside ◇; ◇ inside ☆

TACHS-A3-037

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The three arrangements share the same rule.

1: Top-left ○; top-right □; bottom-left □; bottom-right ○

2: Top-left △; top-right ◇; bottom-left ◇; bottom-right △

3: Top-left ☆; top-right +; bottom-left +; bottom-right ☆

Which choice follows the same rule?

- A) Top-left ○; top-right □; bottom-left ○; bottom-right □
 B) Top-left △; top-right △; bottom-left ◇; bottom-right ◇
 C) Top-left +; top-right ×; bottom-left +; bottom-right ×
 D) Top-left ●; top-right ■; bottom-left ■; bottom-right ●

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R3-026 - Correct: A - Correct Answer: The lantern's latch is bent. - Explanation: Mina seeks repair because the bent latch prevents the glass door from closing.

TACHS-R3-027 - Correct: C - Correct Answer: Careless force can leave lasting damage. - Explanation: Mr. Bell warns that striking the brass could create additional damage.

TACHS-R3-028 - Correct: B - Correct Answer: limited but adequate. - Explanation: The lantern lights only a small area, yet that light is sufficient to reveal stones.

TACHS-R3-029 - Correct: D - Correct Answer: Patience can protect useful things and help people serve others well. - Explanation: Careful repair preserves the lantern, and Mina later uses its light thoughtfully.

TACHS-R3-030 - Correct: A - Correct Answer: Blue marks painted on rocks - Explanation: Regularly spaced blue marks guide the cousins upstream toward the footbridge.

TACHS-R3-031 - Correct: B - Correct Answer: He is uncertain whether the route is approved for hikers. - Explanation: Eli waits for the ranger's confirmation before labeling the route an official temporary detour.

TACHS-R3-032 - Correct: C - Correct Answer: By presenting a problem that experience helps Eli reconsider - Explanation: The blocked trail forces Eli to discover that maps may require careful updates.

TACHS-R3-033 - Correct: D - Correct Answer: Thoughtful - Explanation: The passage calmly reflects on how direct experience changes Eli's understanding.

TACHS-R3-034 - Correct: A - Correct Answer: They loosen soil and tolerate changing moisture levels. - Explanation: The third paragraph directly identifies both benefits of deep-rooted native plants.

TACHS-R3-035 - Correct: C - Correct Answer: place excessive demand on. - Explanation: A sudden large volume of runoff can overwhelm or strain drainage systems.

TACHS-R3-036 - Correct: B - Correct Answer: Rain gardens use carefully chosen sites, soils, and plants to slow and filter runoff, though they have limits. - Explanation: This choice includes the passage's method, design requirements, benefits, and limitations.

TACHS-R3-037 - Correct: D - Correct Answer: To emphasize their combined ability to absorb and slow water in many locations - Explanation: The sponge comparison illustrates the distributed gardens' collective effect on runoff.

TACHS-R3-038 - Correct: A - Correct Answer: To explain how rain gardens work and identify both their benefits and limitations - Explanation: The author objectively describes their operation, placement, value, and limits.

WRITTEN EXPRESSION ANSWERS

TACHS-WE3-027 - Correct: C - Correct Answer: We recorded the sounds with microphones. - Explanation: This revision expresses the original idea directly without redundant references to audible sounds.

TACHS-WE3-028 - Correct: A - Correct Answer: Sentence 7 - Explanation: The cafeteria menu is unrelated to creating the neighborhood audio map.

TACHS-WE3-029 - Correct: B - Correct Answer: After students labeled each clip with its location and time, they rerecorded interviews in which wind had covered voices. - Explanation: The revision clearly subordinates the first action and connects both completed tasks.

TACHS-WE3-030 - Correct: D - Correct Answer: the location with the loudest activity to the location with the least - Explanation: The revision states the intended comparison precisely and consistently.

TACHS-WE3-031 - Correct: B - Correct Answer: Volunteers sorted packets by plant type; they also added labels showing planting depth and season. - Explanation: A semicolon correctly joins the two closely related independent clauses.

TACHS-WE3-032 - Correct: A - Correct Answer: After sentence 5 - Explanation: The example of long-grown plants directly supports the claim about preserving locally successful varieties.

TACHS-WE3-033 - Correct: A - Correct Answer: Though modest in size, the seed library grows stronger with every properly returned envelope. - Explanation: This revision is concise, clear, and appropriately emphasizes the effect of returned seeds.

TACHS-WE3-034 - Correct: D - Correct Answer: Every one of the volunteers receives a name tag. - Explanation: The singular subject "every one" correctly takes the singular verb "receives."

TACHS-WE3-035 - Correct: B - Correct Answer: Between you and me, the final clue was difficult. - Explanation: The preposition "between" requires the objective pronouns "you" and "me."

TACHS-WE3-036 - Correct: C - Correct Answer: The guide taught us how to measure, record, and compare the samples. - Explanation: All three coordinated actions use parallel base-form verbs.

TACHS-WE3-037 - Correct: C - Correct Answer: Line 3 - Explanation: The adverb "attentively," not the adjective "attentive," must modify "listened."

TACHS-WE3-038 - Correct: D - Correct Answer: Line 4 - Explanation: The singular determiner "every" requires "volunteer," not the plural "volunteers."

MATHEMATICS ANSWERS

TACHS-M3-026 - Correct: C - Correct Answer: \$48 - Explanation: A 25% discount is \$16, so the sale price is \$64 - \$16 = \$48.

TACHS-M3-027 - Correct: A - Correct Answer: 19/20 - Explanation: Divide first: $7/10 \div 2 = 7/20$; then $12/20 + 7/20 = 19/20$.

TACHS-M3-028 - Correct: D - Correct Answer: 6 - Explanation: Oats are 2 of 7 ratio parts, so $2/7 \times 21 = 6$ cups.

TACHS-M3-029 - Correct: B - Correct Answer: Multiply by 2, then add 1. - Explanation: Applying this rule produces 9, 19, and 39 successively.

TACHS-M3-030 - Correct: C - Correct Answer: $x = 6$ - Explanation: Expanding gives $8x - 12 = 3x + 18$, so $5x = 30$ and $x = 6$.

TACHS-M3-031 - Correct: D - Correct Answer: $7 + 2m$ - Explanation: The fixed monthly charge is 7, and m rentals cost $2m$.

TACHS-M3-032 - Correct: A - Correct Answer: 24 - Explanation: The values total 120, and $120 \div 5 = 24$.

TACHS-M3-033 - Correct: C - Correct Answer: $1/25$ - Explanation: With replacement, the probability is $2/10 \times 2/10 = 4/100 = 1/25$.

TACHS-M3-034 - Correct: B - Correct Answer: 65° - Explanation: Triangle angles total 180° , so $180 - 47 - 68 = 65^\circ$.

TACHS-M3-035 - Correct: A - Correct Answer: 40 cm - Explanation: The width is $96 \div 12 = 8$, so the perimeter is $2(12 + 8) = 40$.

TACHS-M3-036 - Correct: D - Correct Answer: 3,500 - Explanation: Since 1 liter equals 1,000 milliliters, $3.5 \times 1,000 = 3,500$.

TACHS-M3-037 - Correct: B - Correct Answer: 2 hours 31 minutes - Explanation: From 10:47 to 12:47 is two hours, plus 31 minutes to 1:18.

ABILITY ANSWERS

TACHS-A3-025 - Correct: B - Correct Answer: ■ - Explanation: Across each row, the open shape changes to its filled counterpart without changing form.

TACHS-A3-026 - Correct: A - Correct Answer: ↑ - Explanation: Moving right or down rotates each arrow 90° clockwise, so left becomes up.

TACHS-A3-027 - Correct: A - Correct Answer: $\triangle\triangle\triangle$ - Explanation: One matching open shape is added from the first cell to the second in each row.

TACHS-A3-028 - Correct: B - Correct Answer: $\diamond\triangle$ - Explanation: The two shapes exchange left and right positions across each row.

TACHS-A3-029 - Correct: C - Correct Answer: $\triangle\star$ - Explanation: The third cell combines the first and second cells in their original order.

TACHS-A3-030 - Correct: D - Correct Answer: Two holes near the upper-right and lower-right corners - Explanation: Reversing the horizontal fold reflects the hole vertically while preserving its right-side position.

TACHS-A3-031 - Correct: B - Correct Answer: Four holes, one in each quadrant, reflected across both centerlines - Explanation: Two perpendicular folds create four reflections symmetric across both centerlines.

TACHS-A3-032 - Correct: A - Correct Answer: One diamond-shaped opening above the center - Explanation: A triangular cut on the fold reflects into a matching triangle, forming one diamond-shaped opening.

TACHS-A3-033 - Correct: C - Correct Answer: Two holes mirrored across the upper-left-to-lower-right diagonal - Explanation: Reversing the single diagonal fold creates two locations reflected across that diagonal.

TACHS-A3-034 - Correct: B - Correct Answer: $\diamond\blacklozenge\diamond$ - Explanation: Each arrangement has matching open outer shapes surrounding their filled counterpart.

TACHS-A3-035 - Correct: D - Correct Answer: $\leftarrow \uparrow \rightarrow$ - Explanation: Each successive arrow rotates 90° clockwise, so left, up, right follows the shared rule.

TACHS-A3-036 - Correct: C - Correct Answer: $\square$ inside Δ ; Δ inside $\diamond$ - Explanation: Each figure forms a nested chain of three different open shapes.

TACHS-A3-037 - Correct: D - Correct Answer: Top-left $\bullet$; top-right $\blacksquare$; bottom-left $\blacksquare$; bottom-right $\bullet$ - Explanation: Matching symbols occupy opposite corners, producing diagonal symmetry.

TACHS Practice Exam - Exam 3 - Part D
Filename: tachs_full_exam_03_part_d.txt
Exam Family: TACHS
Exam Number: 3
Exam Part: D
Question Count: 50
QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R3-S10

Section: Reading

Stimulus Type: Literary Passage

Title: The Unclaimed Apron

Text:

When the community kitchen closed after Saturday's bread workshop, Nia noticed a blue apron hanging behind the pantry door. Its pocket held a card covered with careful notes about oven temperatures. She assumed the apron belonged to Mr. Han, the instructor, but he had already left.

Nia could have placed it in the lost-and-found bin. Instead, she studied the card. One note mentioned that the old oven's back corner heated fastest. Another warned that its timer rang two minutes late. The information would matter during Sunday's charity bake sale, when volunteers unfamiliar with the kitchen would prepare dozens of loaves.

Nia called Mr. Han. The apron was not his; it belonged to Mrs. Ortiz, who had tested the oven all week. Mrs. Ortiz asked Nia to leave both items with the morning coordinator. Before going home, Nia copied the two most important warnings onto a large sheet and taped it beside the oven controls.

The next morning, volunteers rotated their pans and used a separate timer. By noon, every loaf had baked evenly. Mrs. Ortiz thanked Nia for returning the apron, then smiled at the sign. "You returned more than what you found," she said.

STIMULUS END

TACHS-R3-039

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S10

Prompt:

What was inside the apron pocket?

- A) A list of volunteers
- B) A card with oven notes
- C) A separate timer
- D) A bread recipe

TACHS-R3-040

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S10

Prompt:

Why did Nia make a large sign?

- A) She wanted to advertise another workshop.
- B) She needed to identify the apron's owner.
- C) She hoped to replace the oven's controls.
- D) She wanted all volunteers to see the useful warnings.

TACHS-R3-041

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S10

Prompt:

In the final sentence, Mrs. Ortiz means that Nia “returned more than what you found” because Nia

- A) shared the apron’s useful information with others.
- B) brought extra supplies into the kitchen.
- C) returned several objects from the lost-and-found bin.
- D) taught the volunteers how to bake bread.

TACHS-R3-042

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R3-S10

Prompt:

Which statement best expresses the passage’s central idea?

- A) Old equipment should always be replaced before a large event.
- B) Written directions are more dependable than experienced instructors.
- C) Acting thoughtfully on a small discovery can help an entire group.
- D) Community events succeed only when one person takes charge.

STIMULUS START

Stimulus ID: TACHS-R3-S11

Section: Reading

Stimulus Type: Scientific Passage

Title: Measuring a Tree Without Climbing It

Text:

Foresters often need to estimate a tree’s height without climbing it. One simple method uses shadows and similar triangles. On a sunny day, a person places a straight measuring stick upright on level ground near the tree. The stick and tree cast shadows at the same time. Because sunlight reaches both objects at the same angle, the two objects and their shadows form triangles with matching angles.

The forester measures the stick, its shadow, and the tree’s shadow. Suppose a one-meter stick casts a two-meter shadow while a tree casts a thirty-meter shadow. The tree is fifteen times taller than the stick because its shadow is fifteen times longer. Its estimated height is therefore fifteen meters.

Accuracy depends on careful conditions. The stick must stand vertically, and both shadows should fall across ground with a similar slope. Blurred shadow tips can make endpoints difficult to judge. Wind may also move branches, changing the apparent end of the tree’s shadow. Taking several measurements and averaging the results can reduce the effect of small errors.

This method does not provide a perfect measurement, but it allows foresters to gather useful information quickly with ordinary tools.

STIMULUS END

TACHS-R3-043

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S11

Prompt:

Why do the tree and stick form triangles with matching angles?

- A) They stand on exactly the same spot.
- B) Their shadows always have equal lengths.
- C) Sunlight reaches both at the same angle.
- D) Foresters measure them with the same tool.

TACHS-R3-044

Section: Reading

Type: mcq
Category: Reading
Skill: Organization, Text Structure, or Literary Technique
Stimulus ID: TACHS-R3-S11
Prompt:
How is the passage mainly organized?
A) It presents a claim and then argues against it.
B) It explains a method, gives an example, and discusses sources of error.
C) It compares two professions and recommends one.
D) It lists events in the history of forest measurement.

TACHS-R3-045
Section: Reading
Type: mcq
Category: Reading
Skill: Evaluative Comprehension, Author's Purpose, or Tone
Stimulus ID: TACHS-R3-S11
Prompt:
Why does the author mention taking several measurements?
A) To explain how averaging can reduce small measurement errors
B) To prove that one measurement can never be useful
C) To suggest that different trees cast identical shadows
D) To show that the stick should be moved after every reading

TACHS-R3-046
Section: Reading
Type: mcq
Category: Reading
Skill: Main Idea, Central Thought, or Summary
Stimulus ID: TACHS-R3-S11
Prompt:
Which statement best summarizes the passage?
A) Tree height can be found exactly whenever the sun is visible.
B) Foresters prefer shadows because measuring tools are unreliable.
C) Wind is the main reason trees are difficult to measure.
D) Shadow proportions offer a practical estimate when measurements are taken carefully.

STIMULUS START
Stimulus ID: TACHS-R3-S12
Section: Reading
Stimulus Type: Biographical-Style Passage
Title: Designing for Many Hands
Text:

As a young craft worker, Elena Varga repaired household tools in a busy neighborhood shop. She noticed that many customers struggled with ordinary objects for reasons manufacturers had overlooked. A jar opener might require a strong grip. A measuring cup might have markings visible from only one side. Instead of blaming users, Varga began asking how the objects could change.

Years later, as a product designer, she invited people of different ages and abilities to test early models. Their comments often altered her plans. When testers had difficulty turning a narrow handle, she made it wider and added a textured surface. When left-handed users covered measurement labels while pouring, she printed labels on both sides.

Some colleagues worried that these changes would make products complicated. Varga argued that thoughtful features could simplify an object for nearly everyone. A broad handle helped a person with limited grip strength, but it also helped a cook whose hands were wet. Labels on both sides assisted left-handed users and anyone viewing the cup from an unusual angle.

Varga's greatest contribution was not a single invention. It was her habit of treating difficulty as information. By observing where people hesitated, she discovered opportunities to design objects that welcomed more users.

STIMULUS END

TACHS-R3-047

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R3-S12

Prompt:

What change did Varga make after testers struggled with a narrow handle?

- A) She removed the handle.
- B) She added measurement labels.
- C) She made it wider and textured.
- D) She replaced the early model with a jar.

TACHS-R3-048

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R3-S12

Prompt:

Which belief most clearly guided Varga's work?

- A) Products should be tested only by trained designers.
- B) A user's difficulty can reveal a weakness in a design.
- C) Specialized products should serve only one group.
- D) Attractive objects are usually the easiest to use.

TACHS-R3-049

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R3-S12

Prompt:

In the final paragraph, "welcomed" most nearly means

- A) greeted aloud.
- B) entertained briefly.
- C) sold directly to.
- D) made usable and inviting for.

TACHS-R3-050

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R3-S12

Prompt:

What is the author's primary purpose?

- A) To show how attentive observation helped a designer create more usable objects
- B) To explain why handmade tools are superior to manufactured ones
- C) To argue that all products should have identical handles
- D) To describe the steps for becoming a professional craft worker

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE3-S07

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Hallway Exhibition

Text:

[1] Our art club planned an exhibition for the long hallway outside the auditorium. [2] Because students hurry through that space between classes. [3] We decided that each display needed to communicate its idea quickly. [4] Members selected twelve pieces and wrote brief labels explaining the materials and inspiration behind them. [5] The labels were printed in large type so that visitors could read them from a comfortable distance. [6] The auditorium curtains are dark red.

[7] On installation day, we arranged the works at consistent heights and checked that no frame blocked a doorway. [8] By afternoon, the hallway had become an orderly gallery that students could enjoy without slowing traffic.

STIMULUS END

TACHS-WE3-039

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S07

Prompt:

Which revision best combines sentences 2 and 3 into one complete sentence?

- A) Because students hurry through that space between classes, we decided that each display needed to communicate its idea quickly.
- B) Students hurry through that space between classes, because deciding each display needed to communicate quickly.
- C) Because students hurry through that space between classes; each display needed to communicate its idea quickly.
- D) Students hurrying through that space between classes and each display communicating quickly.

TACHS-WE3-040

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S07

Prompt:

Which sentence should be removed because it does not support the passage's focus?

- A) Sentence 3
- B) Sentence 4
- C) Sentence 6
- D) Sentence 7

TACHS-WE3-041

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S07

Prompt:

Which revision of sentence 4 is most concise while preserving its meaning?

- A) Members made a selection of twelve different pieces and then proceeded to write labels that were brief.
- B) Twelve pieces were selected by members, who wrote labels about various things concerning them.
- C) Members selected pieces, and the pieces had materials as well as inspiration that labels explained.
- D) Members chose twelve works and wrote brief labels about their materials and inspiration.

TACHS-WE3-042

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE3-S07

Prompt:

Which sentence would best follow sentence 5?

- A) Several club members also enjoy painting landscapes at home.
- B) We also spaced the works apart so viewers could distinguish one display from the next.
- C) The school auditorium hosts concerts during the winter.
- D) Some printed materials are produced by large machines.

STIMULUS START

Stimulus ID: TACHS-WE3-S08

Section: Written Expression
Stimulus Type: Revision Passage
Title: Testing the Solar Cooker
Text:

[1] Our science team built a solar cooker from a box, reflective panels, and a clear cover. [2] We wanted to learn how the angle of the panels affected the temperature inside the box. [3] At ten-minute intervals, Jamal recorded the temperature, checking the position of the sun. [4] The first trial reached only 38°C. [5] For the second trial, we adjusted the panels so more sunlight was directed through the cover. [6] The temperature then rose to 51°C under the same outdoor conditions. [7] The second arrangement was more better at concentrating sunlight. [8] Our results showed that careful panel placement can substantially improve the cooker's performance.

STIMULUS END

TACHS-WE3-043

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE3-S08

Prompt:

Which revision best corrects the misplaced modifier in sentence 3?

- A) Checking the sun's position, the temperature was recorded by Jamal at ten-minute intervals.
- B) Jamal, at ten-minute intervals checking, recorded the temperature and the sun's position.
- C) At ten-minute intervals, Jamal checked the sun's position and recorded the temperature.
- D) The sun's position checked, at ten-minute intervals Jamal recorded the temperature.

TACHS-WE3-044

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE3-S08

Prompt:

Which revision correctly replaces "more better" in sentence 7?

- A) best
- B) better
- C) most better
- D) more good

TACHS-WE3-045

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE3-S08

Prompt:

Which revision of sentence 6 most clearly emphasizes the comparison between trials?

- A) The temperature was 51°C, and outdoor conditions existed.
- B) Under the same outdoor conditions, the second trial reached 51°C, thirteen degrees higher than the first.
- C) It then rose under conditions that were the same outside to a temperature of 51°C.
- D) The second temperature, which was a temperature, rose to 51°C.

TACHS-WE3-046

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses verb tense correctly?

- A) By the time the bus arrived, we had waited for twenty minutes.
- B) By the time the bus arrived, we wait for twenty minutes.
- C) By the time the bus arrives yesterday, we had waited for twenty minutes.

D) By the time the bus arrived, we are waiting for twenty minutes.

TACHS-WE3-047

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses pronouns correctly?

- A) The coach spoke privately with him and me.
- B) Her and Elena designed the poster.
- C) The award was presented to Marcus and I.
- D) Us students carried the boxes inside.

TACHS-WE3-048

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

- Line 1: The collection of old photographs was stored safely.
Line 2: Each image had been placed in a protective sleeve.
Line 3: A volunteer carefully recorded its date and source.
Line 4: Later, the curator organize the images by neighborhood.
- A) Line 1
 - B) Line 2
 - C) Line 3
 - D) Line 4

TACHS-WE3-049

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

- Line 1: The two routes lead to the same overlook.
Line 2: This trail is shorter than the path beside the lake.
Line 3: However, it climbs most steeply than the other route.
Line 4: Both paths are clearly marked.
- A) Line 1
 - B) Line 2
 - C) Line 3
 - D) Line 4

TACHS-WE3-050

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which line contains an error in standard written US English?

- Line 1: Before sunrise, the crew loaded the equipment.
Line 2: They secured every container with a strap.
Line 3: The driver inspected the tires and mirrors.
Line 4: Then the trucks moves slowly toward the gate.
- A) Line 1
 - B) Line 2
 - C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M3-038

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A tank is $\frac{3}{5}$ full. After 18 liters are added, it is $\frac{9}{10}$ full. What is the tank's total capacity?

A) 45 liters

B) 54 liters

C) 72 liters

D) 60 liters

TACHS-M3-039

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $4.8 \div 0.6 + 3 \times 2$?

A) 11

B) 14

C) 16

D) 22

TACHS-M3-040

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A club's membership increased from 80 students to 92 students. What was the percent increase?

A) 15%

B) 12%

C) 18%

D) 20%

TACHS-M3-041

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The perimeter of a rectangle is 46 centimeters. Its length is 5 centimeters more than its width.

What is the width?

A) 8 cm

B) 9 cm

C) 13 cm

D) 18 cm

TACHS-M3-042

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A machine produces 7 finished parts every 12 minutes. At the same constant rate, how many parts does it produce in 60 minutes?

- A) 28
- B) 30
- C) 35
- D) 42

TACHS-M3-043

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The numbers of tickets sold on four days were 36, 42, 39, and 47. How many tickets must be sold on the fifth day for the five-day mean to be 42?

- A) 42
- B) 44
- C) 48
- D) 46

TACHS-M3-044

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A bag contains 4 red, 5 blue, and 3 green tiles. One tile is selected, not replaced, and then a second tile is selected. What is the probability that both tiles are green?

- A) $\frac{1}{11}$
- B) $\frac{1}{12}$
- C) $\frac{1}{22}$
- D) $\frac{1}{44}$

TACHS-M3-045

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

Data Set P is 6, 8, 10, 12, 14. Data Set Q is 2, 8, 10, 12, 18. Which statement is true?

- A) Both sets have the same mean, but Q has a greater range.
- B) P has a greater mean and a greater range.
- C) Q has a greater mean, but both sets have the same range.
- D) Both sets have the same mean and the same range.

TACHS-M3-046

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Each exterior angle of a regular polygon measures 45° . How many sides does the polygon have?

- A) 8
- B) 7
- C) 6
- D) 9

TACHS-M3-047

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A circle has a radius of 6 centimeters. Using $\pi = 3.14$, what is its area?

- A) 37.68 cm^2
- B) 75.36 cm^2
- C) 113.04 cm^2
- D) 226.08 cm^2

TACHS-M3-048

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangular prism measures 8 cm by 5 cm by 3 cm. What is its total surface area?

- A) 79 cm^2
- B) 120 cm^2
- C) 240 cm^2
- D) 158 cm^2

TACHS-M3-049

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A cyclist travels 18 miles in 1.5 hours. At the same rate, how far will the cyclist travel in 2 hours?

- A) 20 miles
- B) 22 miles
- C) 24 miles
- D) 27 miles

TACHS-M3-050

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A recipe requires 250 milliliters of broth per batch. How many liters are needed for 6 batches?

- A) 0.15 L
- B) 1.5 L
- C) 15 L
- D) 150 L

SECTION 4 - ABILITY

TACHS-A3-038

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: ○ above □

Top-right: □ above ○

Bottom-left: ● above ■

Bottom-right: ?

- A) ● above ■
- B) ■ above ●
- C) □ above ○
- D) ■ above ○

TACHS-A3-039

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: ○

Top-right: ○+

Bottom-left: □□

Bottom-right: ?

- A) □+
- B) □□
- C) □□+
- D) +□□+

TACHS-A3-040

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 3×3 matrix.

Row 1: ○ | □ | □□

Row 2: △ | ☆ | △☆

Row 3: ◇ | + | ?

- A) ◇+
- B) +◇
- C) ◇◇
- D) ++

TACHS-A3-041

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the 2×2 matrix.

Top-left: Top ○; Bottom ●

Top-right: Top ●; Bottom ○

Bottom-left: Left □; Right ■

Bottom-right: ?

- A) Left □; Right ■
- B) Top ■; Bottom □
- C) Left ○; Right ●
- D) Left ■; Right □

TACHS-A3-042

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded in half from left to right. The folded rectangle is then folded from bottom to top. One hole is punched near the folded shape's upper-right corner. What appears when

the sheet is fully unfolded?

- A) Two holes near the right corners
- B) Four holes, one near each corner
- C) Four holes along the vertical centerline
- D) Two holes along the top edge

TACHS-A3-043

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded once along the diagonal from upper-left to lower-right. A small triangular cut is made on the folded diagonal, below the center. What appears after unfolding?

- A) One diamond-shaped opening below the center
- B) Two separate triangular holes near the left edge
- C) Four triangular holes around the center
- D) One triangular opening above the center

TACHS-A3-044

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular sheet is folded from right to left along its vertical centerline. One circular hole is punched exactly on the folded sheet's horizontal centerline, near its new right edge. What appears after unfolding?

- A) One hole at the center
- B) Two holes mirrored across the horizontal centerline
- C) Two holes mirrored across the vertical centerline
- D) Four holes mirrored across both centerlines

TACHS-A3-045

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square sheet is folded from top to bottom, then from right to left. A semicircular cut is made on the folded shape's bottom edge, away from all corners and fold intersections. What appears after unfolding?

- A) Two circular openings
- B) Four semicircular notches on the outer edges
- C) Two diamond-shaped openings
- D) Two matching circular openings reflected across the vertical centerline

TACHS-A3-046

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

Three arrangements follow the same rule:

○ | ●

□ | ■

△ | ▲

Which choice follows the same rule?

- A) ◇ | ◆
- B) ◆ | ◇

- C) ☆ | ○
D) ★ | ★

TACHS-A3-047

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

Three arrangements follow the same rule:

↑ → ↓

→ ↓ ←

↓ ← ↑

Which choice follows the same rule?

A) ← ↓ →

B) ↑ ← ↓

C) ← ↑ →

D) → ↑ ←

TACHS-A3-048

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

Three arrangements follow the same rule:

○ inside □; ● outside □

△ inside ◇; ▲ outside ◇

☆ inside ○; ★ outside ○

Which choice follows the same rule?

A) ◆ inside △; ◇ outside △

B) □ inside ☆; ■ outside ☆

C) ■ inside ○; □ outside ○

D) ○ inside ○; ● outside □

TACHS-A3-049

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

Three arrangements follow the same rule:

Top: ○; Center: □; Bottom: ○

Top: △; Center: ☆; Bottom: △

Top: ◇; Center: +; Bottom: ◇

Which choice follows the same rule?

A) Top: ○; Center: ●; Bottom: □

B) Top: +; Center: ○; Bottom: ×

C) Top: ☆; Center: ☆; Bottom: +

D) Top: ■; Center: □; Bottom: ■

TACHS-A3-050

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

Three groups follow the same rule:

○ □ △ ○

□ △ ○ □

△ ○ □ △

Which choice follows the same rule?

A) ○ □ ○ △

B) □ □ △ □

C) △ □ ○ △

D) ☆ + × ☆

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R3-039 - Correct: B - Correct Answer: A card with oven notes - Explanation: The passage states that the apron pocket contained a card of notes about oven temperatures.

TACHS-R3-040 - Correct: D - Correct Answer: She wanted all volunteers to see the useful warnings. - Explanation: Nia enlarged the two warnings so Sunday's unfamiliar volunteers could use them.

TACHS-R3-041 - Correct: A - Correct Answer: shared the apron's useful information with others. - Explanation: Nia returned the apron and also made its valuable information available to the entire group.

TACHS-R3-042 - Correct: C - Correct Answer: Acting thoughtfully on a small discovery can help an entire group. - Explanation: Nia's careful response to one found object prevents baking problems for many volunteers.

TACHS-R3-043 - Correct: C - Correct Answer: Sunlight reaches both at the same angle. -

Explanation: Equal sunlight angles produce similar triangles for the stick and tree.

TACHS-R3-044 - Correct: B - Correct Answer: It explains a method, gives an example, and discusses sources of error. - Explanation: The passage progresses from procedure to numerical example and then to accuracy concerns.

TACHS-R3-045 - Correct: A - Correct Answer: To explain how averaging can reduce small measurement errors - Explanation: The author directly identifies repeated measurements and averaging as ways to limit minor errors.

TACHS-R3-046 - Correct: D - Correct Answer: Shadow proportions offer a practical estimate when measurements are taken carefully. - Explanation: This choice captures both the method's usefulness and its need for careful conditions.

TACHS-R3-047 - Correct: C - Correct Answer: She made it wider and textured. - Explanation: Varga widened the narrow handle and added texture after observing testers.

TACHS-R3-048 - Correct: B - Correct Answer: A user's difficulty can reveal a weakness in a design. - Explanation: Varga treats hesitation as evidence that an object may need improvement.

TACHS-R3-049 - Correct: D - Correct Answer: made usable and inviting for. - Explanation: The objects welcomed more users by becoming easier and more comfortable for them to use.

TACHS-R3-050 - Correct: A - Correct Answer: To show how attentive observation helped a designer create more usable objects - Explanation: The biography emphasizes Varga's observation-based approach and its benefits for varied users.

WRITTEN EXPRESSION ANSWERS

TACHS-WE3-039 - Correct: A - Correct Answer: Because students hurry through that space between classes, we decided that each display needed to communicate its idea quickly. - Explanation: The dependent clause is correctly joined to an independent clause with a comma.

TACHS-WE3-040 - Correct: C - Correct Answer: Sentence 6 - Explanation: The curtain color does not support the passage's focus on planning and installing the hallway exhibition.

TACHS-WE3-041 - Correct: D - Correct Answer: Members chose twelve works and wrote brief labels about their materials and inspiration. - Explanation: This revision preserves all essential information while eliminating unnecessary wording.

TACHS-WE3-042 - Correct: B - Correct Answer: We also spaced the works apart so viewers could distinguish one display from the next. - Explanation: The sentence adds a relevant design choice that supports clear viewing.

TACHS-WE3-043 - Correct: C - Correct Answer: At ten-minute intervals, Jamal checked the sun's position and recorded the temperature. - Explanation: Jamal clearly performs both actions, eliminating the misplaced modifier.

TACHS-WE3-044 - Correct: B - Correct Answer: better - Explanation: The standard comparative form of "good" is "better," without "more."

TACHS-WE3-045 - Correct: B - Correct Answer: Under the same outdoor conditions, the second trial reached 51°C, thirteen degrees higher than the first. - Explanation: The revision states the controlled condition and quantifies the difference clearly.

TACHS-WE3-046 - Correct: A - Correct Answer: By the time the bus arrived, we had waited for twenty minutes. - Explanation: The past perfect correctly describes waiting that occurred before another past event.

TACHS-WE3-047 - Correct: A - Correct Answer: The coach spoke privately with him and me. -

Explanation: The preposition "with" requires the objective pronouns "him" and "me."
 TACHS-WE3-048 - Correct: D - Correct Answer: Line 4 - Explanation: The past-tense context requires "organized," not "organize."
 TACHS-WE3-049 - Correct: C - Correct Answer: Line 3 - Explanation: Comparing two routes requires "more steeply," not "most steeply."
 TACHS-WE3-050 - Correct: D - Correct Answer: Line 4 - Explanation: The plural subject "trucks" requires the plural verb "move."

MATHEMATICS ANSWERS

TACHS-M3-038 - Correct: D - Correct Answer: 60 liters - Explanation: The added 18 liters equal $9/10 - 3/5 = 3/10$ of capacity, so capacity is $18 \div 3/10 = 60$.
 TACHS-M3-039 - Correct: B - Correct Answer: 14 - Explanation: First, $4.8 \div 0.6 = 8$ and $3 \times 2 = 6$; then $8 + 6 = 14$.
 TACHS-M3-040 - Correct: A - Correct Answer: 15% - Explanation: The increase is 12, and $12 \div 80 = 0.15 = 15\%$.
 TACHS-M3-041 - Correct: B - Correct Answer: 9 cm - Explanation: With width w , $2w + 2(w + 5) = 46$, so $4w = 36$ and $w = 9$.
 TACHS-M3-042 - Correct: C - Correct Answer: 35 - Explanation: Sixty minutes contains five 12-minute intervals, so the machine produces $5 \times 7 = 35$ parts.
 TACHS-M3-043 - Correct: D - Correct Answer: 46 - Explanation: Five days must total $5 \times 42 = 210$; the first four total 164, leaving 46.
 TACHS-M3-044 - Correct: C - Correct Answer: $1/22$ - Explanation: Without replacement, the probability is $3/12 \times 2/11 = 6/132 = 1/22$.
 TACHS-M3-045 - Correct: A - Correct Answer: Both sets have the same mean, but Q has a greater range. - Explanation: Both totals are 50, while P's range is 8 and Q's range is 16.
 TACHS-M3-046 - Correct: A - Correct Answer: 8 - Explanation: Exterior angles total 360° , so the polygon has $360 \div 45 = 8$ sides.
 TACHS-M3-047 - Correct: C - Correct Answer: 113.04 cm^2 - Explanation: Area is $\pi r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$.
 TACHS-M3-048 - Correct: D - Correct Answer: 158 cm^2 - Explanation: Surface area is $2(8 \times 5 + 8 \times 3 + 5 \times 3) = 2(79) = 158$.
 TACHS-M3-049 - Correct: C - Correct Answer: 24 miles - Explanation: The rate is $18 \div 1.5 = 12$ miles per hour, and $12 \times 2 = 24$.
 TACHS-M3-050 - Correct: B - Correct Answer: 1.5 L - Explanation: Six batches need $6 \times 250 = 1,500$ milliliters, which equals 1.5 liters.

ABILITY ANSWERS

TACHS-A3-038 - Correct: B - Correct Answer: ■ above ● - Explanation: Moving right swaps top and bottom positions; moving down changes each open shape to its filled counterpart.
 TACHS-A3-039 - Correct: C - Correct Answer: □+ - Explanation: Moving right adds a plus sign, while moving down changes one circle into two squares.
 TACHS-A3-040 - Correct: A - Correct Answer: ◇+ - Explanation: In each row, the third cell combines the first and second symbols in the same order.
 TACHS-A3-041 - Correct: D - Correct Answer: Left ■; Right □ - Explanation: Across each row, the two symbols exchange positions while keeping their forms and fill states.
 TACHS-A3-042 - Correct: B - Correct Answer: Four holes, one near each corner - Explanation: Reversing perpendicular folds reflects the punch across both centerlines, producing one hole near each corner.
 TACHS-A3-043 - Correct: A - Correct Answer: One diamond-shaped opening below the center - Explanation: The cut lies on the fold, so its reflected triangular half forms one diamond-shaped opening.
 TACHS-A3-044 - Correct: C - Correct Answer: Two holes mirrored across the vertical centerline - Explanation: One vertical fold creates two horizontally reflected holes; the centerline position prevents additional vertical separation.
 TACHS-A3-045 - Correct: D - Correct Answer: Two matching circular openings reflected across the vertical centerline - Explanation: The cut on the horizontal fold becomes one circle, and reversing the vertical fold creates its mirror.
 TACHS-A3-046 - Correct: A - Correct Answer: ◇ | ◆ - Explanation: Each valid pair shows an open shape followed by the filled form of the same shape.
 TACHS-A3-047 - Correct: C - Correct Answer: ← ↑ → - Explanation: Each arrow rotates 90° clockwise from the one before it.
 TACHS-A3-048 - Correct: B - Correct Answer: □ inside ☆; ■ outside ☆ - Explanation: Each group places an open shape inside a boundary and its filled counterpart outside that same boundary.
 TACHS-A3-049 - Correct: D - Correct Answer: Top: ■; Center: □; Bottom: ■ - Explanation: Each valid

arrangement has identical top and bottom symbols with a different center symbol.
TACHS-A3-050 - Correct: D - Correct Answer: ☆ + × ☆ - Explanation: Each valid group repeats its first symbol in fourth position while using two different middle symbols.